NOVA SCHOOL OF BUSINESS & ECONOMICS

Economics of education

4th class

Readings

- Hanushek et al., (2011) Chapter 2 – Nonproduction Benefits of Education: Crime, Health, and Good Citizenship
- Testa, P. A. (2018). Education and propaganda: Tradeoffs to public education provision in nondemocracies. *Journal of Public Economics*, *160*, 66-81.

Outline for today

- Why do governments want to intervene in education? (Pt 2)

Why do governments want to intervene in education?

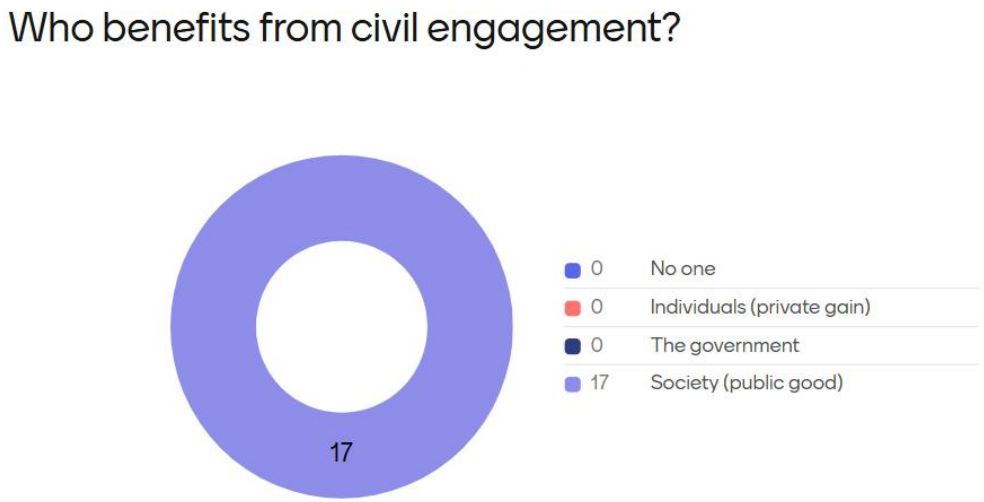
- Empirical evidence for effects of education on **civil participation**
- Cover a model that illustrates the incentives to invest in education by **predatory** vs **benevolent** governments

Why do governments want to intervene in education?

Civil engagement:

a process in which people take collective action to address issues of public concern

Quizz


Who benefits from civil engagement?
17
0 No one
0 Individuals (private gain)
0 The government
17 Society (public good)

Mentimeter
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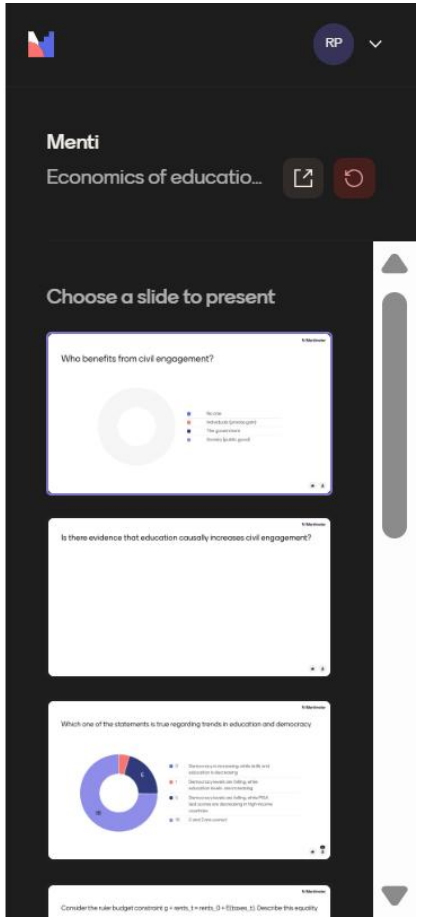
17 of 34 responded


RP
Menti
Economics of educatio...
Choose a slide to present

Why do governments want to intervene in education?

Who benefits from civil engagement?

- Governments? It is not on the government best interest to be scrutinized
- Private gain? Not really
- Society as a whole (public good)

Why do governments want to intervene in education?

Who benefits from civil engagement?

- Having an engaged & informed society increases accountability for government, institutions and businesses
- Increases government scrutiny
- It is also expected to increase democratic values
- It provides a societal benefit, not necessarily a private one

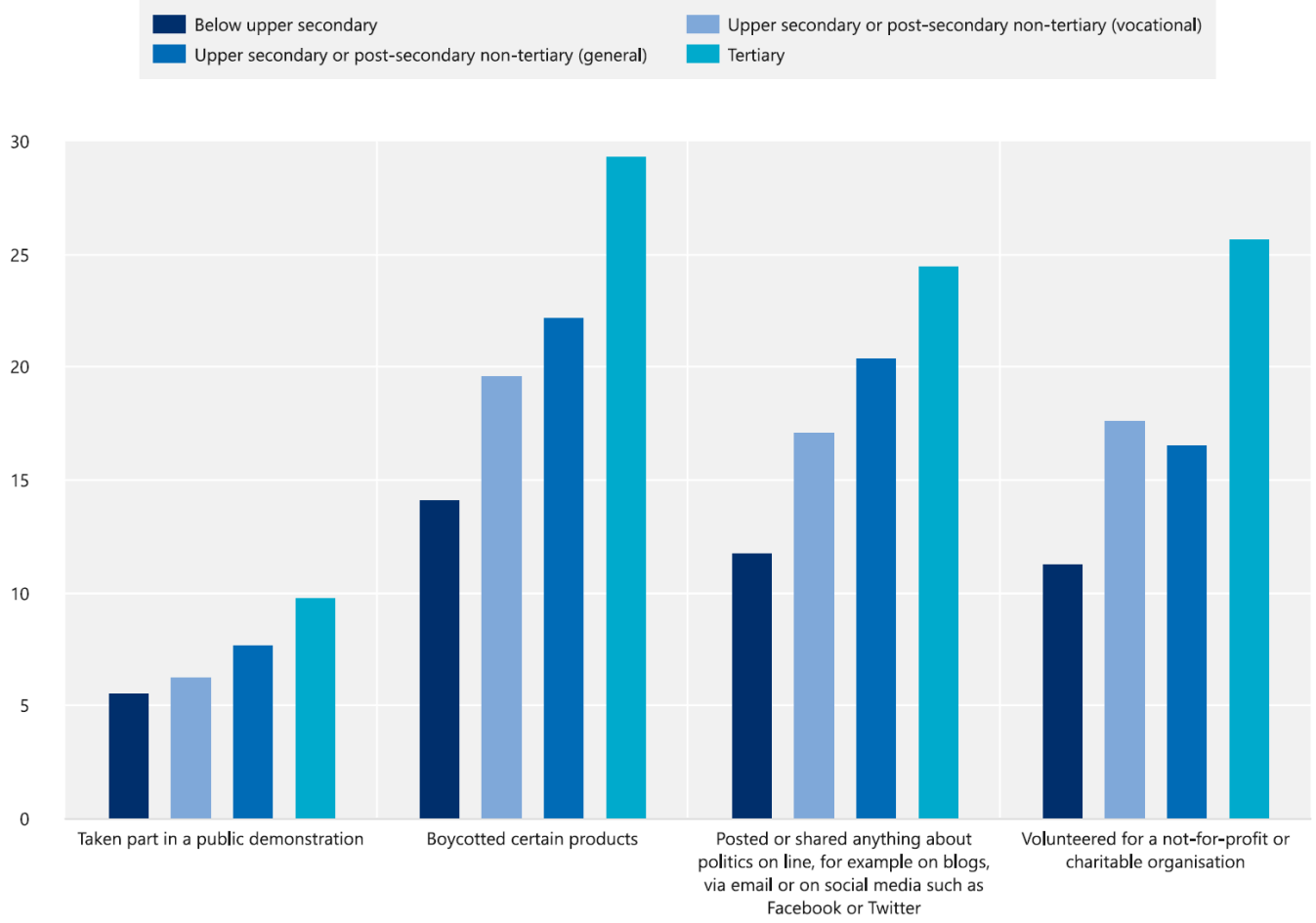
What does the evidence tell us?

Civil engagement

- Correlational evidence

Figure A6.2. Share of adults who reported the following behaviour indicating civic engagement, by educational attainment and programme orientation (2020)

Average of OECD and accession countries participating in the ESS Round 10; 25-64 year-olds; in per cent



Source: OECD (2023), Table A6.2. For more information see *Definitions*, *Methodology* and *Source* sections and *Education at a Glance 2023 Sources, Methodologies and Technical Notes* (OECD, 2023[5]).

Why do governments want to intervene in education?

- Education increases political participation

Table 2

Estimated Effects from an Extra Year of Compulsory Schooling

	[1]	[2]	[3]	[4]
Outcome	*Mean*	*Change in mean from 1 year of compulsory schooling before income controls*	*Change in mean from 1 year of compulsory schooling after income controls*	*Sample specification*
Log weekly income	3.06	0.131 (0.006)***	NA	Working
Log occupational prestige score	3.27	0.063 (0.003)***	0.046 (0.003)***	Working
Unemployed (*share*)	0.045	–0.005 (0.002)***	NA	Full
On welfare (*share*)	0.019	–0.015 (0.002)***	NA	Full
In jail (*share*)	0.027	–0.006 (0.003)*	NA	Black men 21–65 yrs old
In mental institute (share)	0.003	–0.001 (0.000)*	NA	Full
Probability of dying within 10 years[a]	0.11	–0.037 (0.006)***	NA	14 yrs old in 1914–1939 in 1960–1980 U.S. Censuses
First child born when teenager (*share*)	0.078	–0.008 (0.002)***	–0.008 (0.002)***	Women
Divorced (*share*)	0.137	–0.01 (0.002)***	–0.005 (0.003)***	Full
Oldest child behind grade level (*share*)	0.199	–0.032 (0.004)***	–0.026 (0.006)***	Parent matched to oldest 8–16 year old in household
Voted last national election (validated) (*share*)	0.49	0.09 (0.025)***	0.089 (0.027)***	Voting age 1978–2000 November Current Population Survey
Satisfied with life overall (*share*)	0.86	0.048 (0.010)***	0.035 (0.012)***	25–65 year olds born in UK from 1973–98 Eurobarometers

Source: Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic perspectives*, *25*(1), 159-184.

Quizz

Civil engagement

- Evaluate the evidence on education and civil engagement summarized in Hanushek et al., (2011) Chapter 2 (next 3 slides)
- **Question:** Is there evidence that education causally increases civil engagement?
- 5 minutes to read and answer

Hanushek et al., (2011) Chapter 2

What does the evidence tell us?

Table 2.9 Effects of Education on Voter Registration and Voting

Study	Data	Education Measure	Instrumental Variable	Voting/Registration Measure	Sample Mean	IV Estimate	% Effect
A. United States							
Dee (2004)	HS&B	College enroll.	Local 2-yr colleges	Registered to vote	0.67	0.218*	32.5
				Voted in last 12 months	0.36	0.176*	48.9
				Voted in 1988 presidential election	0.55	0.178	32.4
Dee (2004)	GSS	Years of school	State school/ Labor laws	Voted in last presidential election	0.73	0.068*	9.3
Milligan et al. (2004)	NES	High school graduation	State school/ Labor laws	Registered to vote	0.82	0.093	11.3
				Voted in last election	0.68	0.288*	42.3
Milligan et al. (2004)	CPS	High school graduation	State school/ Labor laws	Voted in last election	0.62	0.435*	70.4
B. United Kingdom							
Milligan et al. (2004)	BES	Years of school	National minimum school age	Voted in last election	0.85	–0.008	–0.9
Milligan et al. (2004)	EB	Years of school	National minimum school age	Name on electoral list	0.92	–0.014	–1.5
C. Germany							
Siedler (2007, 2010)	ALLBUS	Years of school	National minimum school age	Voted in last election	0.93[a]	0.005	0.5
Siedler (2007, 2010)	ForsaBus	Years of school	National minimum school age	Voted in last election	0.93[a]	0.000	0.0

Notes: HS&B = High School and Beyond; GSS = General Social Surveys; NES = National Election Survey; CPS = Current Population Survey November Voting Supplements; BES = British Election Studies; EB = Eurobarometer surveys for Britain and Northern Ireland.
[a]Sample means taken from ALLBUS data in Table 3 of Siedler (2007) for those "not affected by the reform."
*Reflects statistically significant at 0.05 level.

Hanushek et al., (2011) Chapter 2

What does the evidence tell us?

Table 2.10 Effects of Education on Political Interest, Information, Views, and Participation

Study	Country	Data	Education Measure	Instrumental Variable	Outcome	Sample Mean	IV Estimate	% Effect
A. Political Interest								
Milligan et al. (2004)	United States	NES	High school graduation	State school/labor laws	Interested in election	0.30	0.270*	90.0
Siedler (2007, 2010)	Germany	ALLBUS	Years of school	National minimum school age	Interested in politics	0.30[a]	–0.096*	–32.0
Siedler (2007, 2010)	Germany	ForsaBus	Years of school	National minimum school age	Interested in politics	0.30[a]	–0.008	–2.7
B. Political Information								
Dee (2004)	United States	GSS	Years of school	State school/labor laws	Newspaper readership index (0–4)	3.2	0.113*	3.5
Milligan et al. (2004)	United States	NES	High school completion	State school/labor laws	Follow campaign on TV	0.79	0.392*	49.6
					Follow campaign in newspapers	0.66	0.852*	129.1
					Follow public affairs	0.66	0.544*	82.4
Milligan et al. (2004)	United States	EB	Years of school	National minimum school age	Follow news every day	0.9	–0.007	–0.8
C. Political Views/Philosophy								
Dee (2004)	United States	GSS	Years of school	State school/labor laws	Allow homosexuals to speak	0.73	0.123*	16.8
					Allow militarist to speak	0.59	0.036*	6.1
Siedler (2007, 2010)	Germany	ALLBUS	Years of school	National minimum school age	Demonstrating is civil right	0.67[a]	–0.001	–0.1

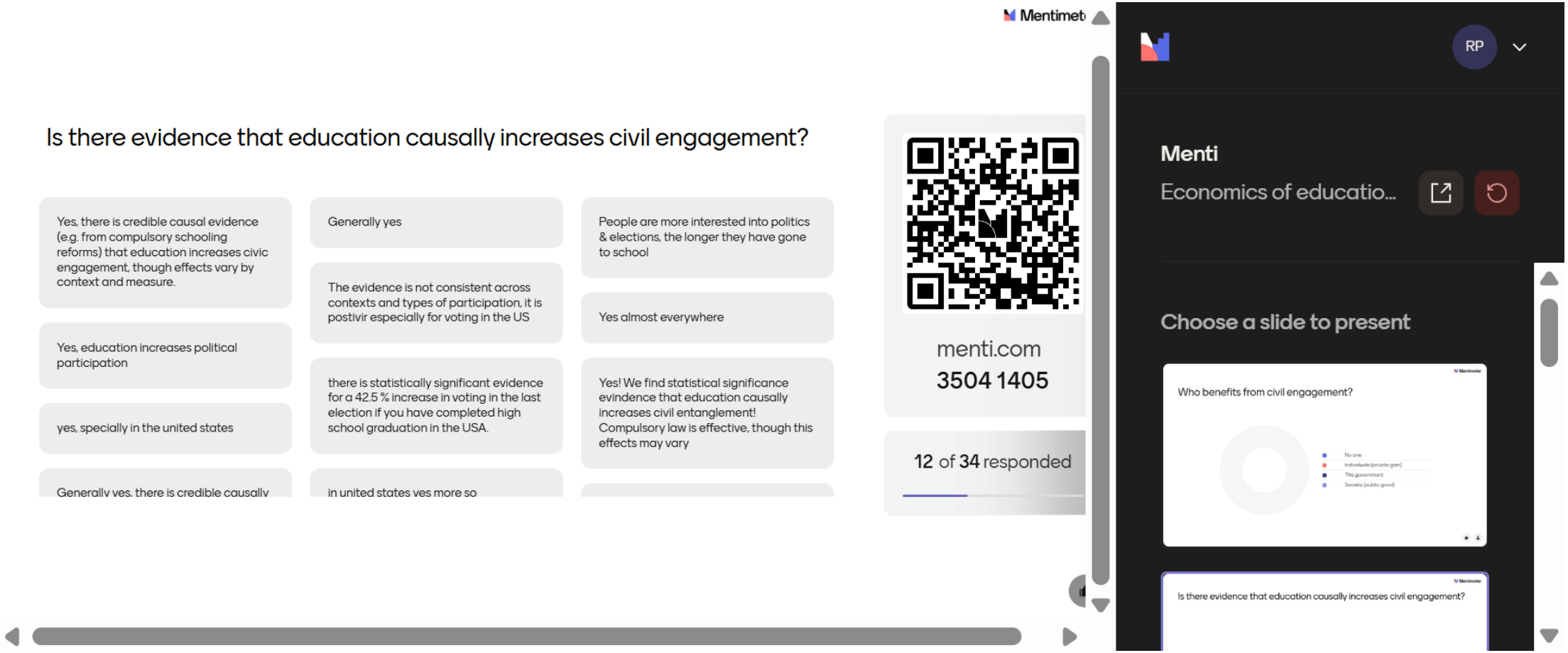
What does the evidence tell us?

Table 2.10 Effects of Education on Political Interest, Information, Views, and Participation—continued

Study	Country	Data	Education Measure	Instrumental Variable	Outcome	Sample Mean	IV Estimate	% Effect
D. Political/Social Participation								
Dee (2004)	United States	HS&B	College enroll.	Local 2-yr colleges	Volunteered in last 12 months	0.37	–0.047	–12.7
Dee (2004)	United States	GSS	Years of school	State school/labor laws	Group memberships	1.8	0.164	9.1
Milligan et al. (2004)	United States	NES	High school grad.	State school/labor laws	Attend political meetings	0.07	0.132	188.6
Milligan et al. (2004)	United Kingdom	EB	Years of school	National minimum school age	Often or occasionally try to persuade others to share views	0.45	0.066*	14.7
					Discuss political matters with friends at least occasionally	0.67	0.095*	14.2
Siedler (2007, 2010)	Germany	ALLBUS	Years of school	National minimum school age	Active in citizen group	0.17[a]	–0.050	–29.4
					Participated in demonstration	0.09[a]	–0.016	–17.8

Notes: HS&B = High School and Beyond; GSS = General Social Surveys; NES = National Election Survey; EB = Eurobarometer surveys for Britain and Northern Ireland.
[a]Sample means taken from ALLBUS data in Table 3 of Siedler (2007) for those "not affected by the reform."
*Reflects statistically significant at 0.05 level.

Quizz


Mentimet
Is there evidence that education causally increases civil engagement?
Yes, there is credible causal evidence (e.g. from compulsory schooling reforms) that education increases civic engagement, though effects vary by context and measure.
Yes, education increases political participation
yes, specially in the united states
Generally yes, there is credible causally
Generally yes
The evidence is not consistent across contexts and types of participation, it is postivir especially for voting in the US
there is statistically significant evidence for a 42.5 % increase in voting in the last election if you have completed high school graduation in the USA.
in united states yes more so
People are more interested into politics & elections, the longer they have gone to school
Yes almost everywhere
Yes! We find statistical significance evindence that education causally increases civil entanglement! Compulsory law is effective, though this effects may vary
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12 of 34 responded
RP
Menti
Economics of educatio...
Choose a slide to present
Who benefits from civil engagement?
No one
Individuals (private gain)
The government
Society (public good)
Is there evidence that education causally increases civil engagement?

Quizz

My response would be:

There is strong evidence that education increases political participation (i.e., voting), interest, information and openness. The effect seems to be particularly present in the US, perhaps driven by relatively lower starting levels of participation. Some evidence of increased civil engagement in the UK as well.

What does the evidence tell us?

Civil engagement

- Is there evidence that education increases democracy?

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2 – Nonproduction Benefits of Education: Crime, Health, and Good Citizenship

What does the evidence tell us?

Civil engagement

- Sample of 1970-2000
- 104 countries

Source: Castelló-Climent, A. (2008). On the distribution of education and democracy. *Journal of Development Economics*, *87*(2), 179-190.

What does the evidence tell us?

Democracy is measured by the Freedom House Political Rights Index

3rd quintile Education: education of the bottom 60%

Source: Castelló-Climent, A. (2008). On the distribution of education and democracy. *Journal of Development Economics*, *87*(2), 179-190.

Table 1
Democracy and education

	Pooled OLS			FE	Dif GMM	System GMM			Different samples. System GMM			
									Dictatorships	Developed	Developing	Non-oil
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Democracy$_{t-5}$	0.702[a]	0.702[a]	0.689[a]	0.399[a]	0.528[a]	0.647[a]	0.651[a]	0.649[a]	0.590[a]	0.592[a]	0.585[a]	0.629[a]
	(0.031)	(0.031)	(0.032)	(0.037)	(0.080)	(0.059)	(0.059)	(0.058)	(0.060)	(0.112)	(0.065)	(0.065)
Average Education$_{t-5}$	0.027[a]		0.010[b]	−0.005	0.006	0.026[a]		0.004	0.000	0.006	−0.013	0.001
	(0.004)		(0.005)	(0.015)	(0.036)	(0.007)		(0.011)	(0.014)	(0.009)	(0.015)	(0.011)
3rd Quintile		0.412[a]	0.295[a]	−0.017	0.059		0.510[a]	0.414[a]	0.479[a]	0.189[b]	0.611[a]	0.458[a]
Education$_{t-5}$		(0.058)	(0.083)	(0.148)	(0.219)		(0.164)	(0.152)	(0.173)	(0.091)	(0.170)	(0.160)
Constant	0.055[b]	0.090[a]	0.074[a]	0.438[a]		0.099[a]	0.093[b]	0.100[b]	0.116[b]	0.261[a]	0.160[a]	0.125[a]
	(0.022)	(0.022)	(0.022)	(0.086)		(0.036)	(0.044)	(0.035)	(0.045)	(0.058)	(0.052)	(0.036)
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.71	0.71	0.72	0.24								
Countries	104	104	104	104	102	104	104	104	73	23	69	92
Observations	764	764	764	764	659	764	746	764	528	179	505	676
AR (2) test					[0.604]	[0.697]	[0.776]	[0.758]	[0.830]	[0.317]	[0.735]	[0.728]
Hansen *J* test					[0.186]	[0.110]	[0.043]	[0.199]	[0.940]	[1.000]	[0.975]	[0.380]
Difference Hansen						[0.501]	[0.023]	[0.405]	[1.000]	[1.000]	[1.000]	[0.615]

Robust standard errors in parenthesis. a, b and c are 1, 5 and 10% significance level respectively. *Democracy* is measured through the political rights index. *Education* includes the average per years of schooling of the population 25 years and over and the percentage of education attained by the 60% of the population. The instruments for column (5)–(12) are the levels of the explanatory variables lagged two periods and further lags until a maximum of 5. In addition to these variables, the system GMM also uses as instruments for the level equations the explanatory variables in the first differences lagged one period.

What does the evidence tell us?

Civil engagement

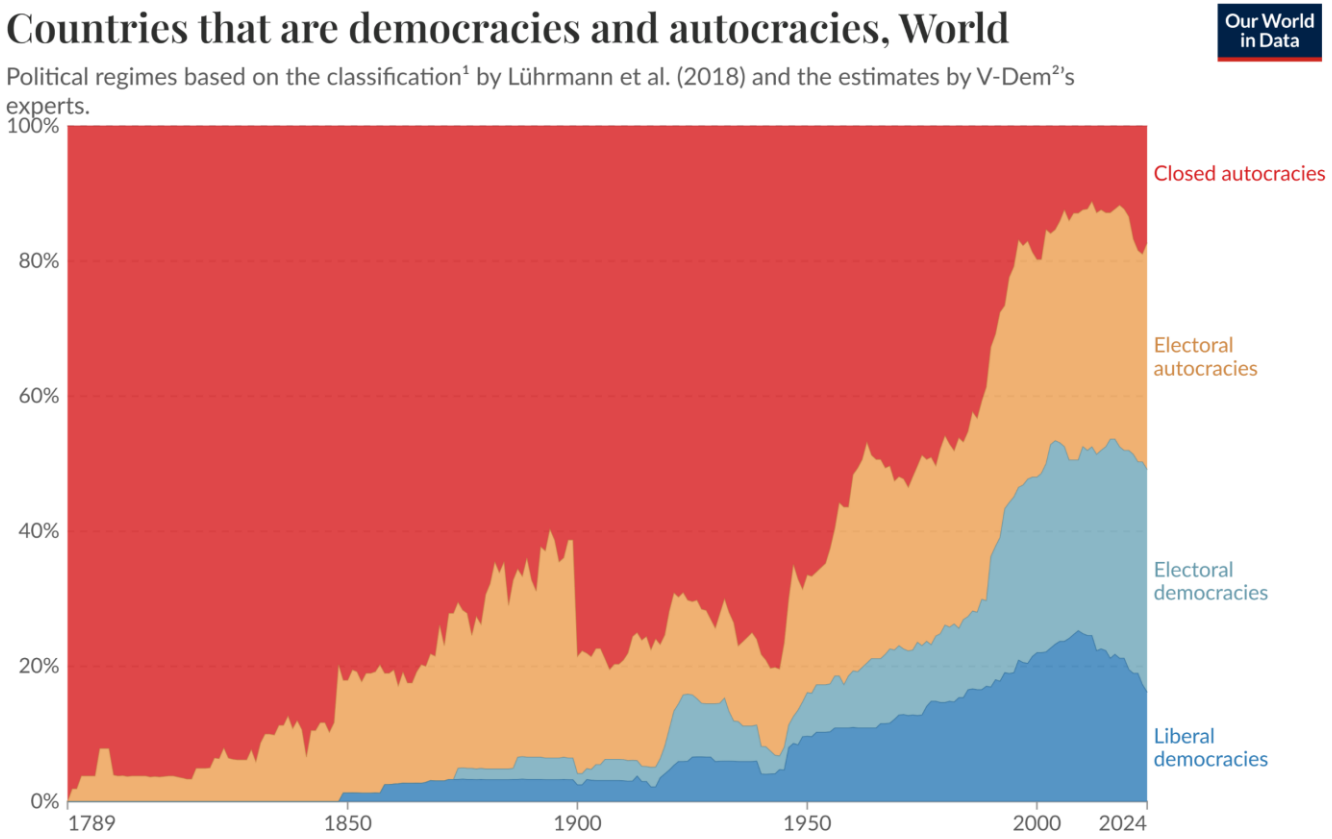
- Authors find that 5 year lagged education predicts Democracy
- More important than average education seems to be the distribution of education
- Authors argue that a large mass of moderately educated citizens seems to be particularly important for democracy

Source: Castelló-Climent, A. (2008). On the distribution of education and democracy. *Journal of Development Economics*, *87*(2), 179-190.

What does the evidence tell us?

Civil engagement

- Democracy has recently been on decline, even in developed, highly educated nations
- Unclear how the relationship between education and democracy has evolved
- Although we did see a fall in PISA tests recently



Data source: V-Dem (2025)

OurWorldinData.org/democracy | CC BY

Note: The share of closed autocracies increases a lot in 1900 because V-Dem covers many more countries since then, often colonies.

1. Regimes of the World's regime classification: Closed autocracy: citizens do not have the right to choose either the chief executive of the government or the legislature through multi-party elections. Electoral autocracy: citizens have the right to choose the chief executive and the legislature through multi-party elections; but they lack some freedoms, such as the freedoms of association or expression that make the elections meaningful, free, and fair. Electoral democracy: citizens have the right to choose the chief executive and the legislature in meaningful, free and fair, and multi-party elections. Liberal democracy: electoral democracy and citizens enjoy individual and minority rights, are equal before the law, and the actions of the executive are constrained by the legislative and the courts. Read more in our article: The 'Regimes of the World' data: how do researchers measure democracy?

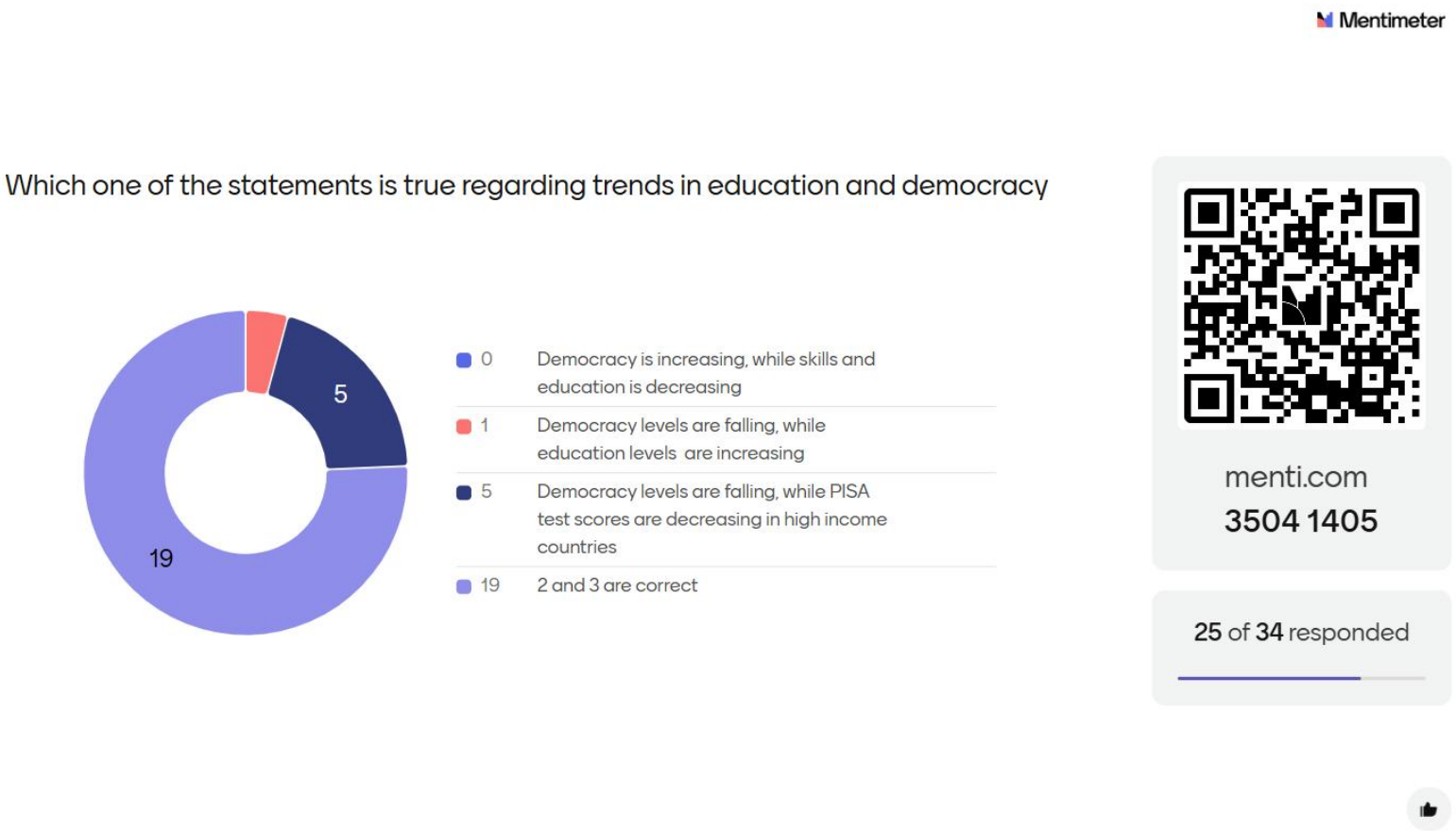
2. V-Dem: The Varieties of Democracy (V-Dem) project publishes data and research on democracy and human rights. It relies on evaluations by around 3,500 country experts and supplementary work by its own researchers to assess political institutions and the protection of rights. The project is managed by the V-Dem Institute, based at the University of Gothenburg in Sweden. Learn more: Democracy data: how do researchers measure democracy? The 'Varieties of Democracy' data: how do researchers measure democracy? The 'Varieties of Democracy' data: how do researchers measure human rights?

What does the evidence tell us?

Civil engagement

- Strong correlation between education and civil engagement
- Micro studies show evidence that education increases civil participation and engagement
- At the macro level, education seems to increase democracy
- Education distribution in the population seems to play an important role

Quizz


Mentimeter
Which one of the statements is true regarding trends in education and democracy
5
19
0 Democracy is increasing, while skills and education is decreasing
1 Democracy levels are falling, while education levels are increasing
5 Democracy levels are falling, while PISA test scores are decreasing in high income countries
19 2 and 3 are correct
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RP
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Economics of educatio...
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Questions? Comments?

Education and civil engagement

Education and propaganda: Tradeoffs to public education provision in nondemocracies☆

Patrick A. Testa

Department of Economics, University of California, Irvine, United States

ARTICLE INFO

JEL classification:
I25
H52
O43

Keywords:
Public education
Propaganda
Political participation
Second-best
Development
Human capital
Social capital
Nondemocracy

ABSTRACT

Nondemocratic regimes face a tradeoff when investing in public education. Education promotes human capital acquisition, expanding the tax base. Yet it also enhances political sophistication and participation, at a cost to nondemocratic regimes. To relax this tradeoff, a regime can disseminate propaganda through its education system. I show that even Bayesian citizens can be influenced by propaganda. By deterring political opposition, propaganda can induce nondemocracies to invest in education when they otherwise would not, improving social welfare. When propaganda is too strong, however, it can generate a backlash. Using cross-country and survey data, I find evidence consistent with the predictions.

Education and civil engagement

This paper proposes a model where:

- Rulers make decisions on **education**, **propaganda**, **taxation**, and **rent extraction**
- Citizens choose whether to receive **education** (if available) and if they receive education, they can choose to join a **political club**
- If they join a political club, they can **protest** government's rents

Education and civil engagement

The model is built on two data-based facts:

- Human capital generates economic growth
- Human capital increases civil engagement and scrutiny
- Although a very simplified version, it is an interesting model to think about the incentives of each actor regarding education

Education and civil engagement

The model:

- Two players: a ruler and a citizen
- Two types of rulers:

$$\theta = \begin{cases} 1 = benevolent\ (maximizes\ social\ welfare) \\ 0 = predatory\ (maximizes\ own\ rents) \end{cases}$$

Education and civil engagement

Ruler chooses

a) whether to invest in public education **gθ = 0 or 1**

b) the curriculum **mθ = 0 or 1** (neutral education or biased with propaganda)

c) how much rent to collect **Rθ**

Citizen chooses

a) Whether to enrol in public education **e = 0 or 1**

b) whether to join a political club **c = 0 or 1**

c) whether to challenge the ruler **r= 0 or 1**

Education and civil engagement

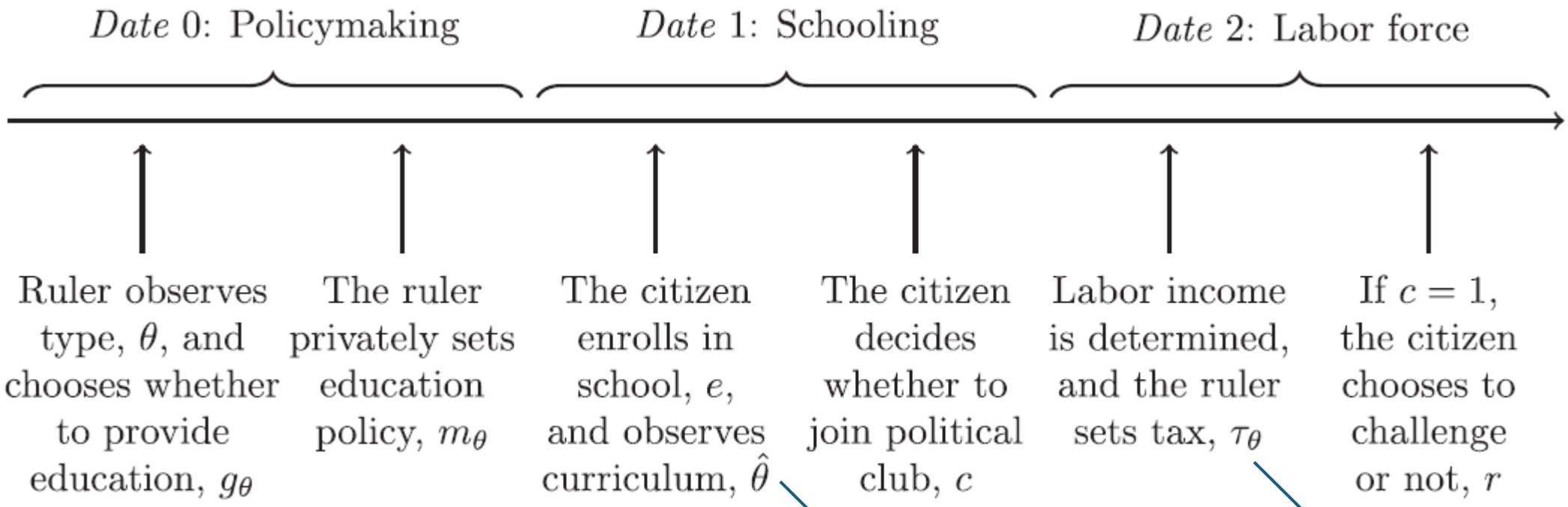
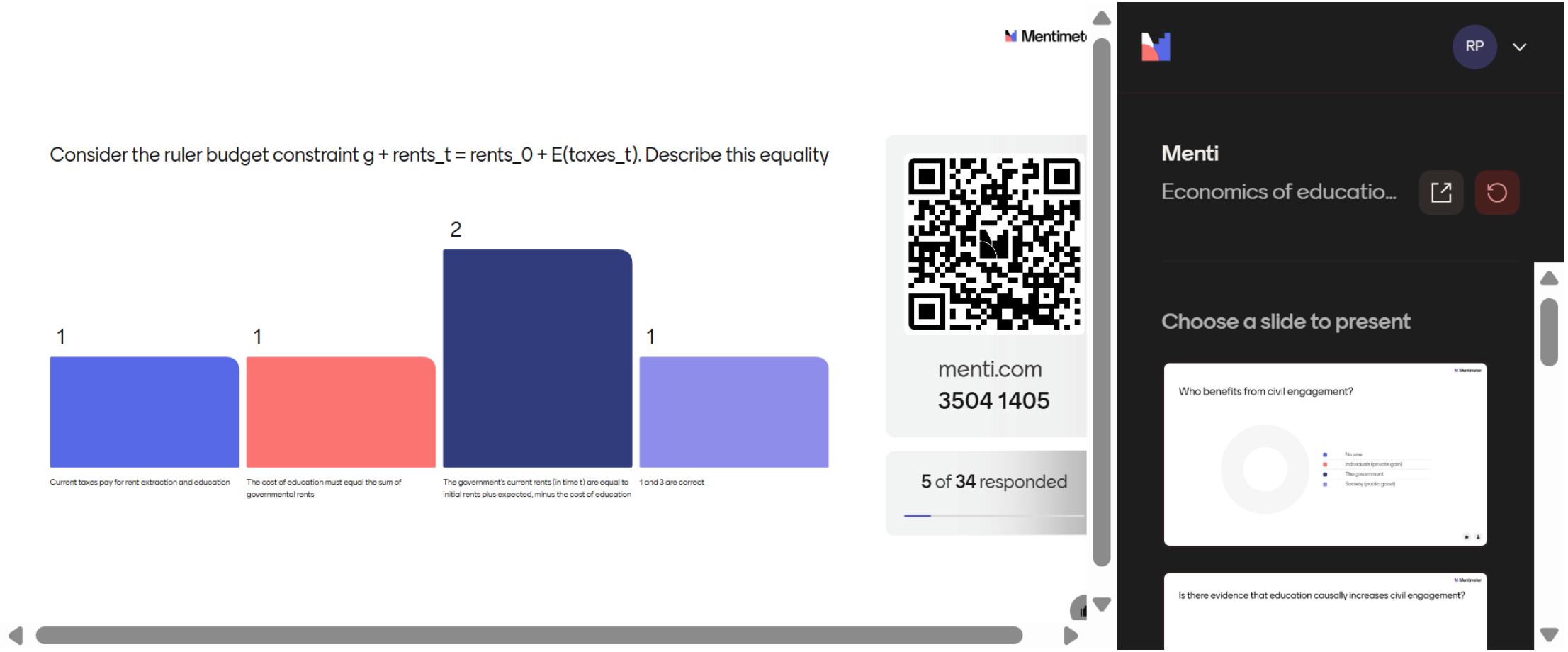


Fig. 1. Timing of actions and events.

Signal on the ruler's type based on the curriculum

g + Rents_t = Rents_0 + E(taxes)

Quizz


Consider the ruler budget constraint g + rents_t = rents_0 + E(taxes_t). Describe this equality
1
1
2
1
Current taxes pay for rent extraction and education
The cost of education must equal the sum of governmental rents
The government's current rents (in time t) are equal to initial rents plus expected, minus the cost of education
1 and 3 are correct
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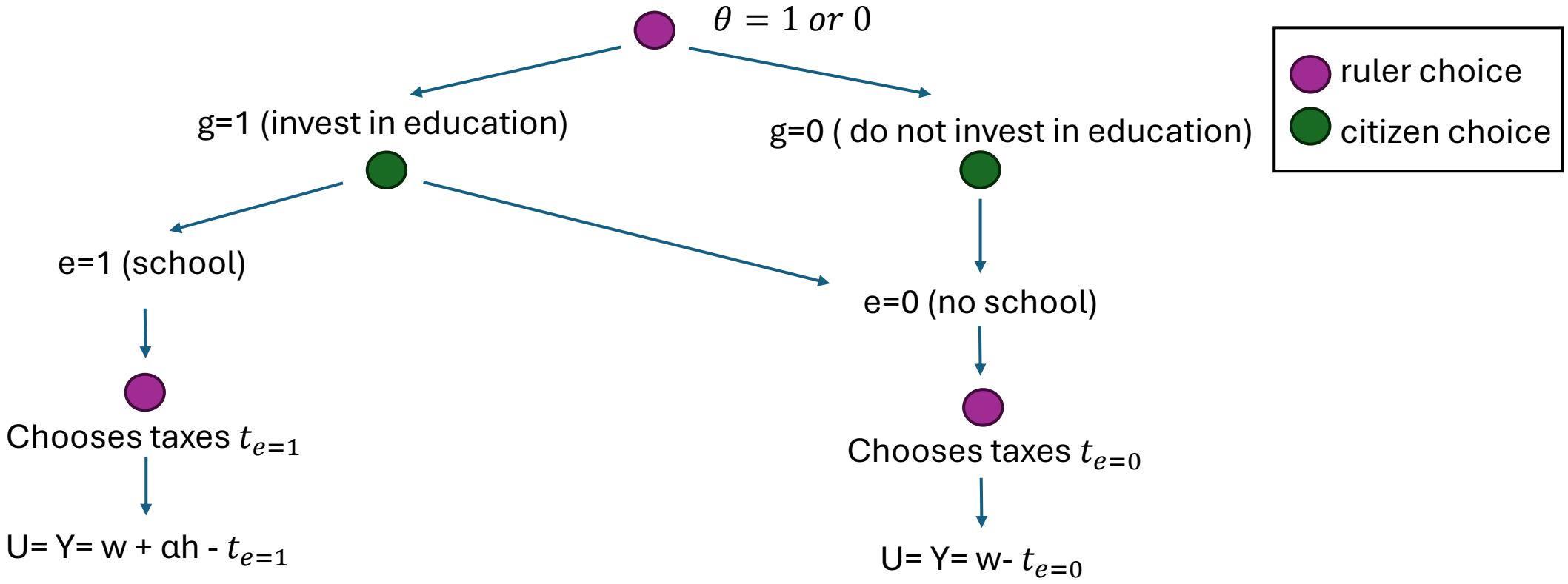
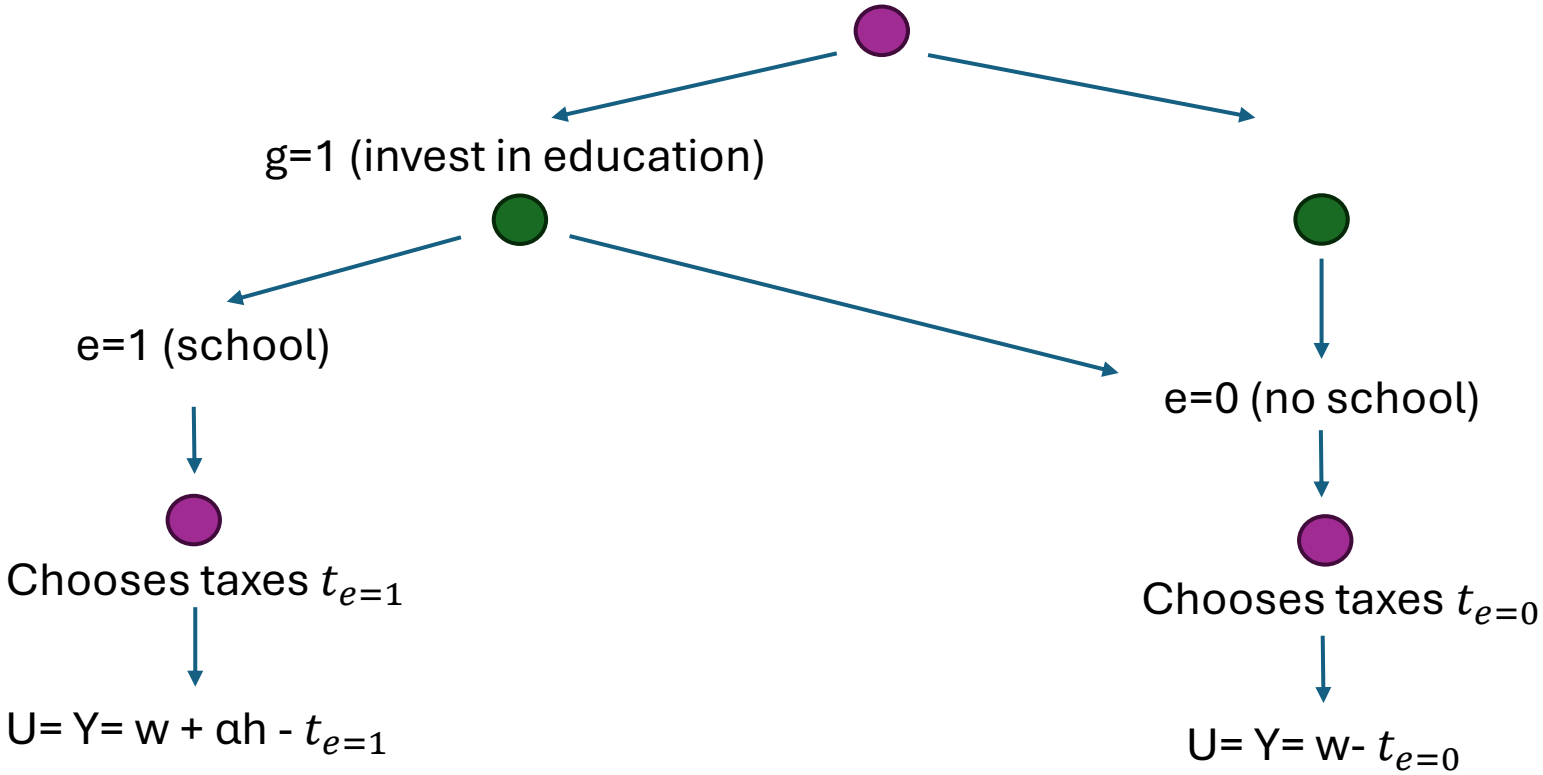
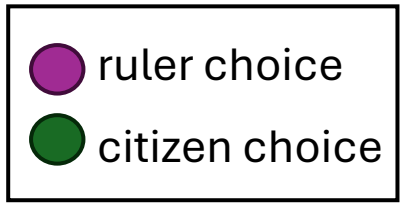
RP
Menti
Economics of educatio...
Choose a slide to present
Who benefits from civil engagement?
Is there evidence that education causally increases civil engagement?

$$\theta = \begin{cases} 1 = benevolent: Max\ U \\ 0 = predatory\ (maximizes\ own\ rents): Max\ R_t = (R_0 + E(t) - g) \end{cases}$$



Education increases Income (Y)
by productivity of human capital
(α) and human capital (h)



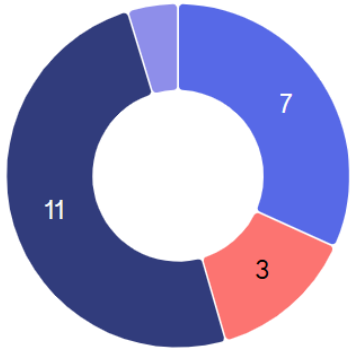


Q: What does a benevolent ruler do in this scenario?
What does a predatory ruler do in this scenario?

Quizz

Mentimeter

What does a benevolent ruler do in this scenario? What about a predatory ruler?



7	Both invest in education, but the predatory will charge maximum amount of taxes possible while the benevolent will charge t=g
3	Benevolent invests in education and charges t=g and the predatory will not invest in education and charges t=g
11	Benevolent invests in education and charges t=g and the predatory will not invest in education and charges t=w
1	None of the above

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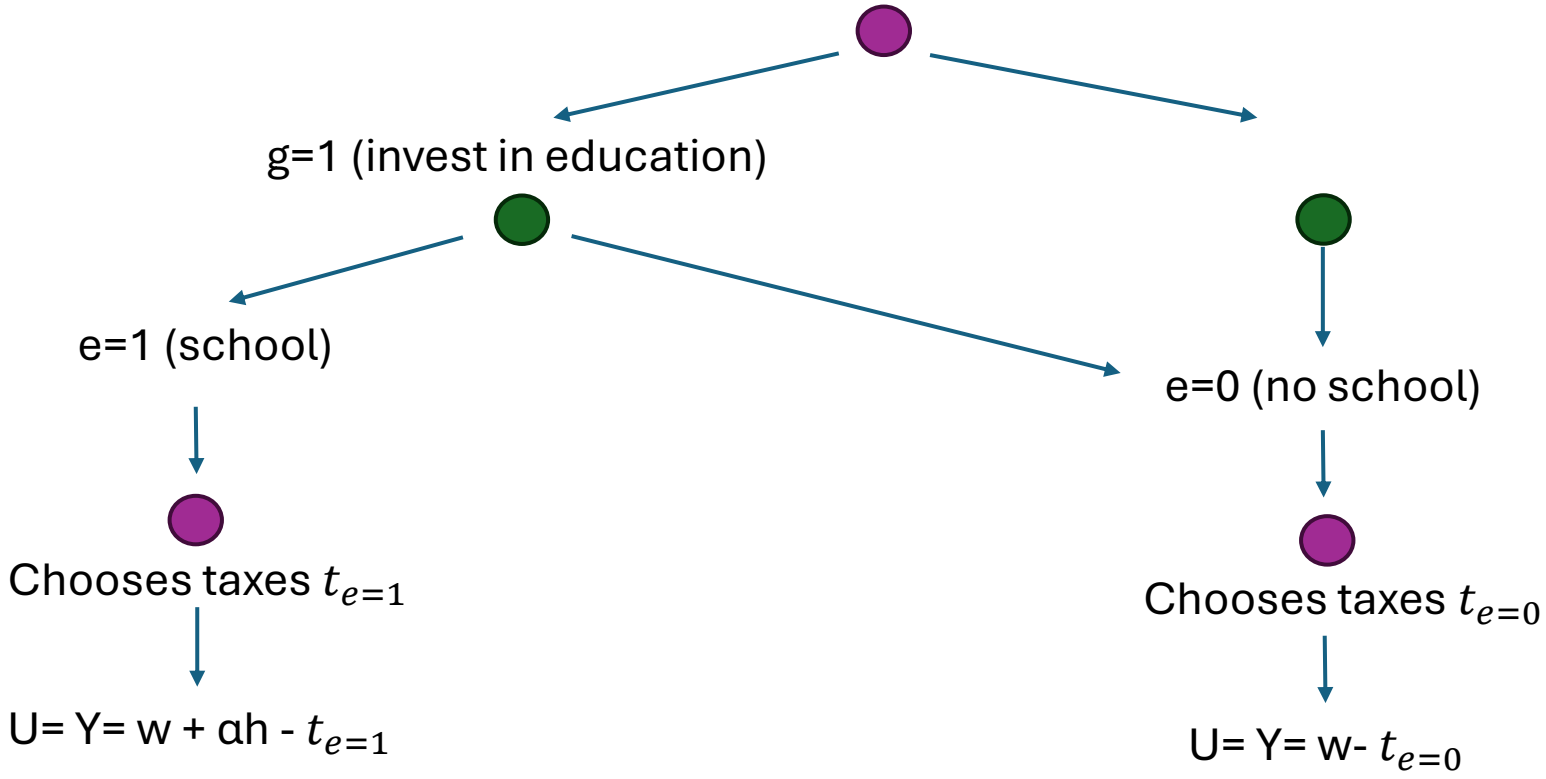
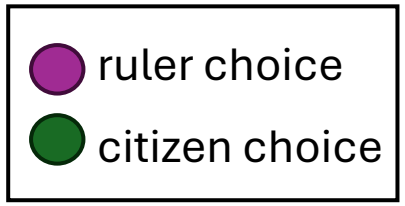
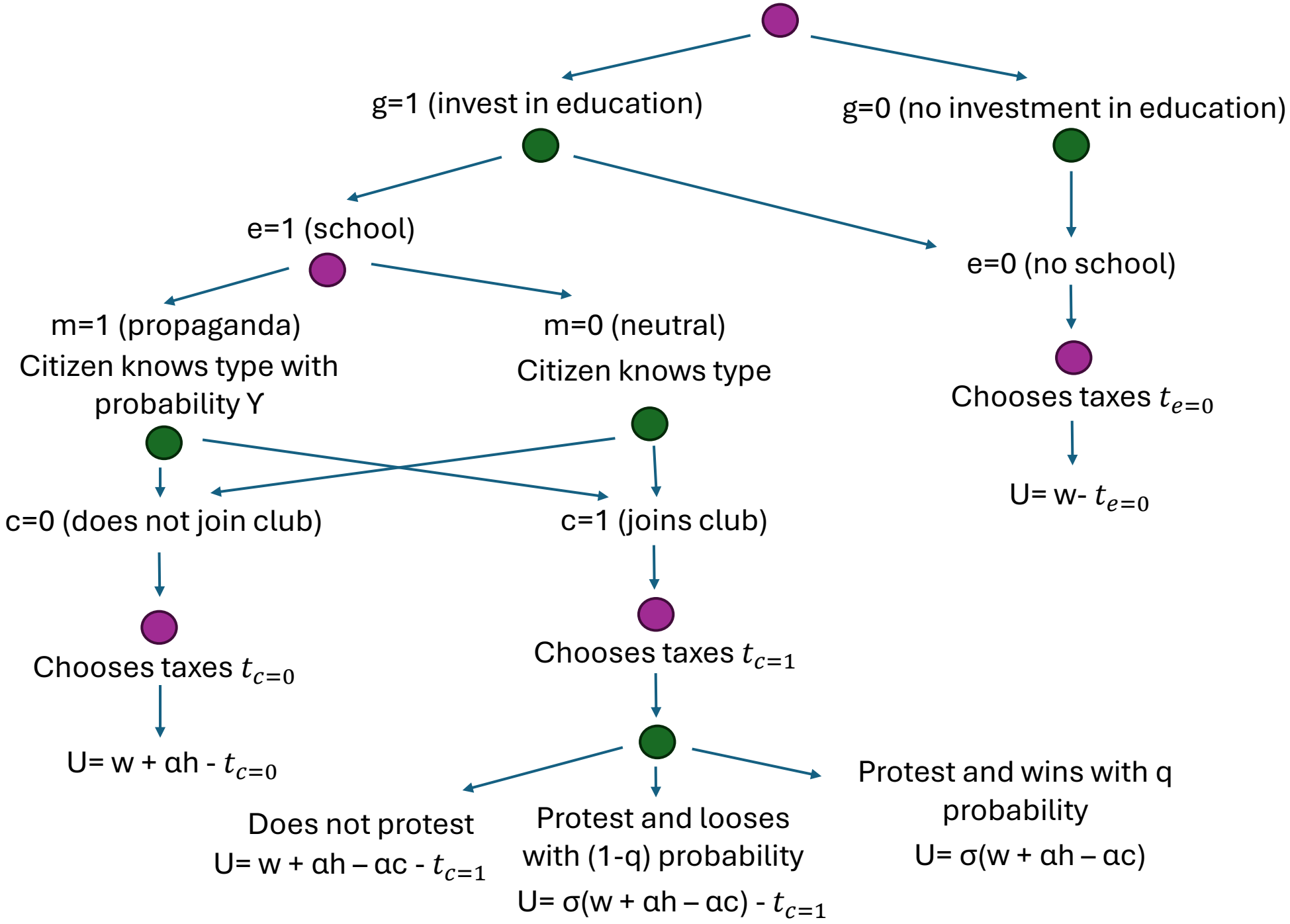
22 of 34 responded





Q: What does a benevolent ruler do in this scenario?
What does a predatory ruler do in this scenario?

A: Benevolent chooses education, and sets $t_{e=1}$ = g
Predatory chooses education, and sets $t_{e=1}$ = w + αh



ruler choice

citizen choice

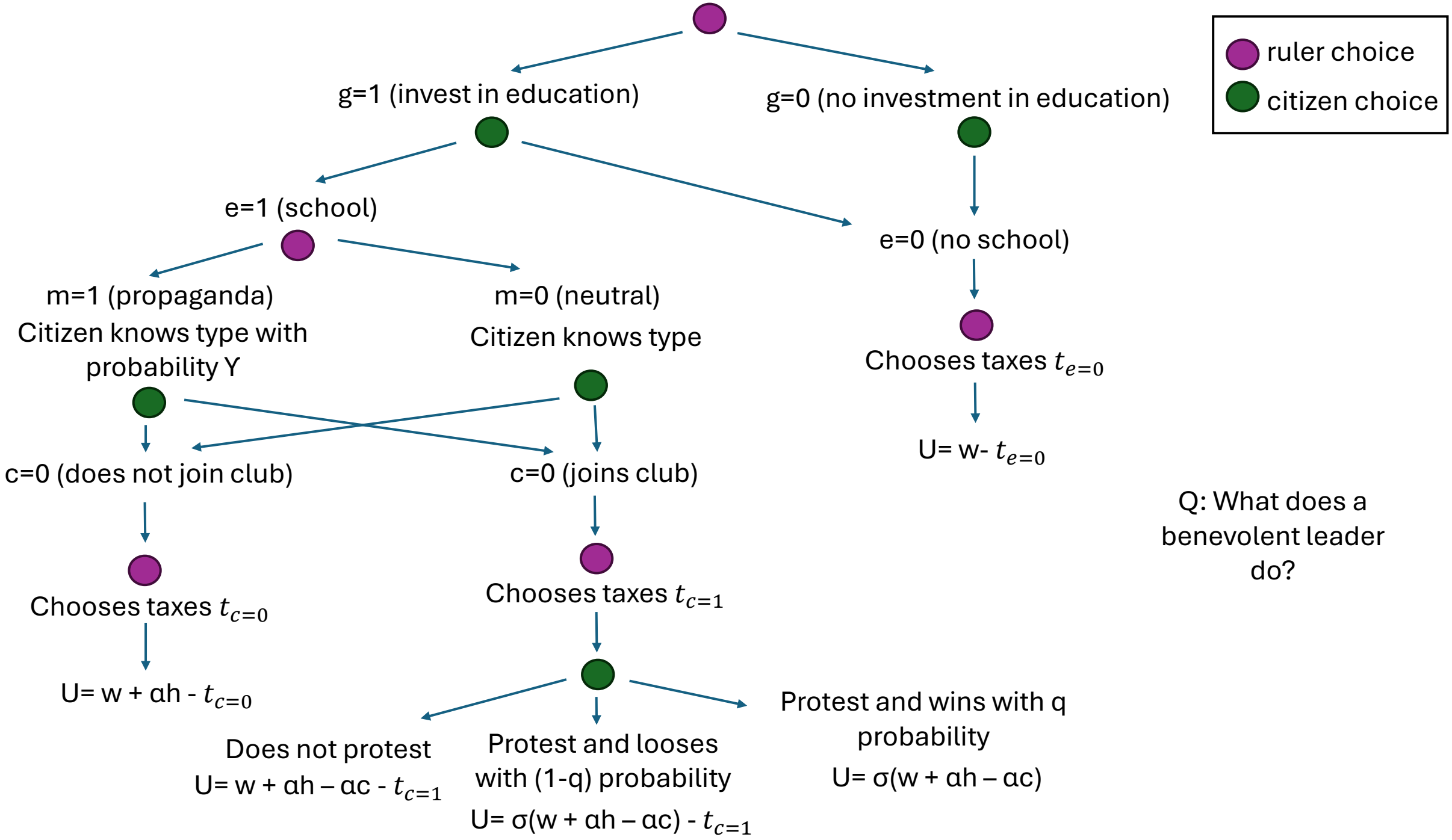
m = propaganda in curriculum

c = joins a political club

ac = cost of joining a political club

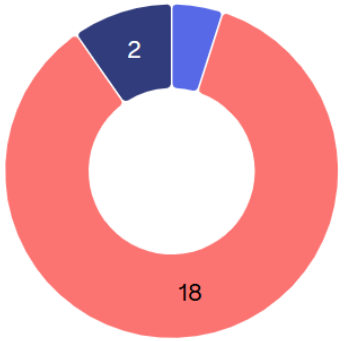
q= probability of winning the protest = political power/repression ability

σ=destruction of income due to protest = rule of law

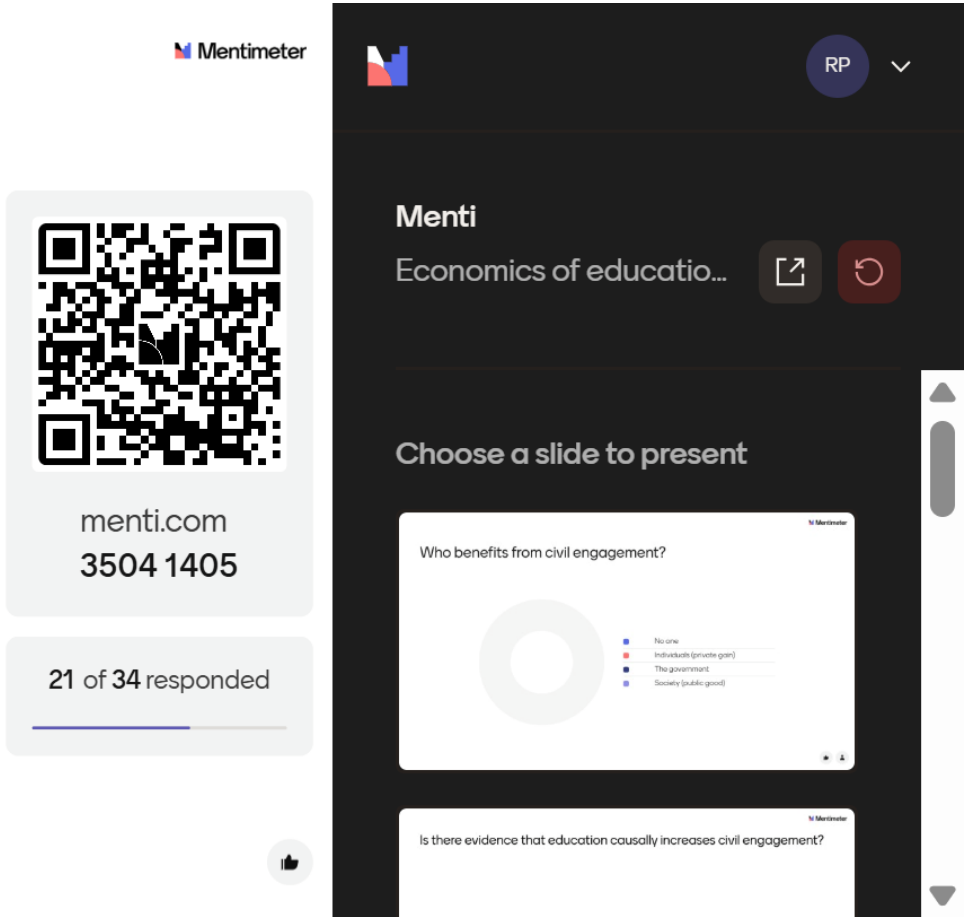

ruler choice
citizen choice
g=1 (invest in education)
g=0 (no investment in education)
e=1 (school)
e=0 (no school)
m=1 (propaganda)
Citizen knows type with probability Y
m=0 (neutral)
Citizen knows type
Chooses taxes $t_{e=0}$
U= w- $t_{e=0}$
c=0 (does not join club)
c=0 (joins club)
Chooses taxes $t_{c=0}$
U= w + αh - $t_{c=0}$
Chooses taxes $t_{c=1}$
Does not protest
U= w + αh – αc - $t_{c=1}$
Protest and looses with (1-q) probability
U= σ(w + αh – αc) - $t_{c=1}$
Protest and wins with q probability
U= σ(w + αh – αc)
Q: What does a benevolent leader do?

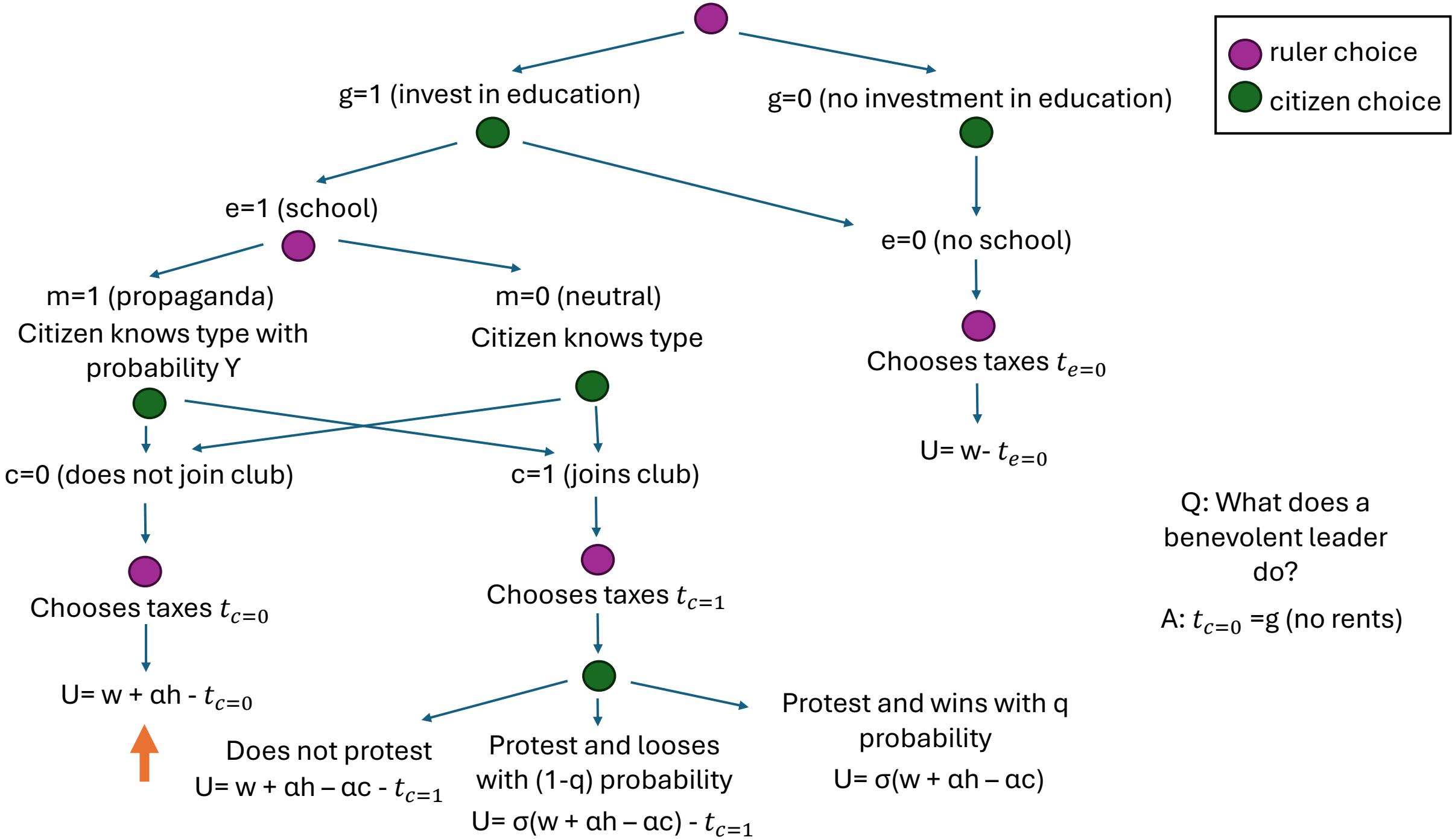
Quizz

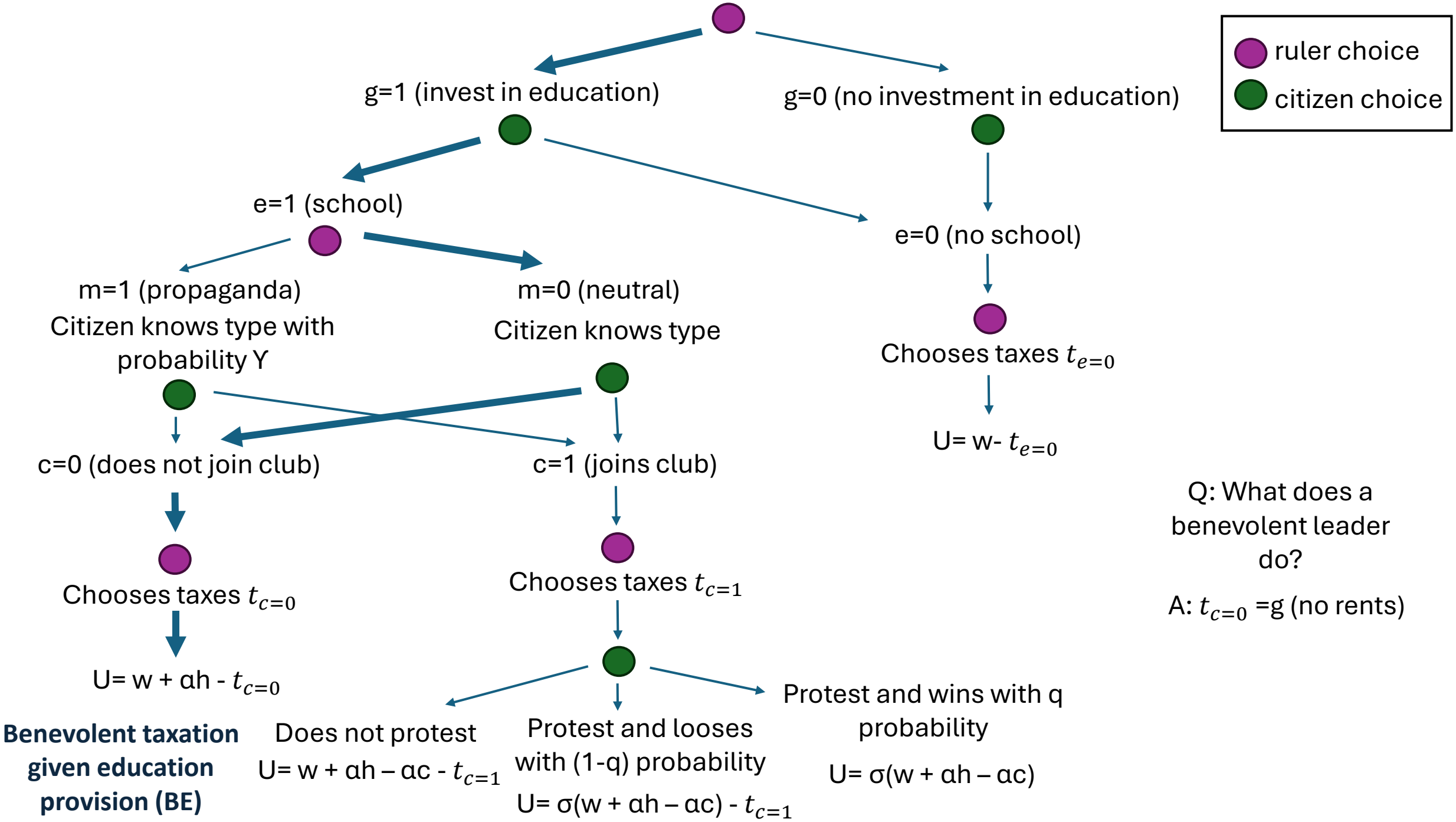
What does a benevolent leader do?

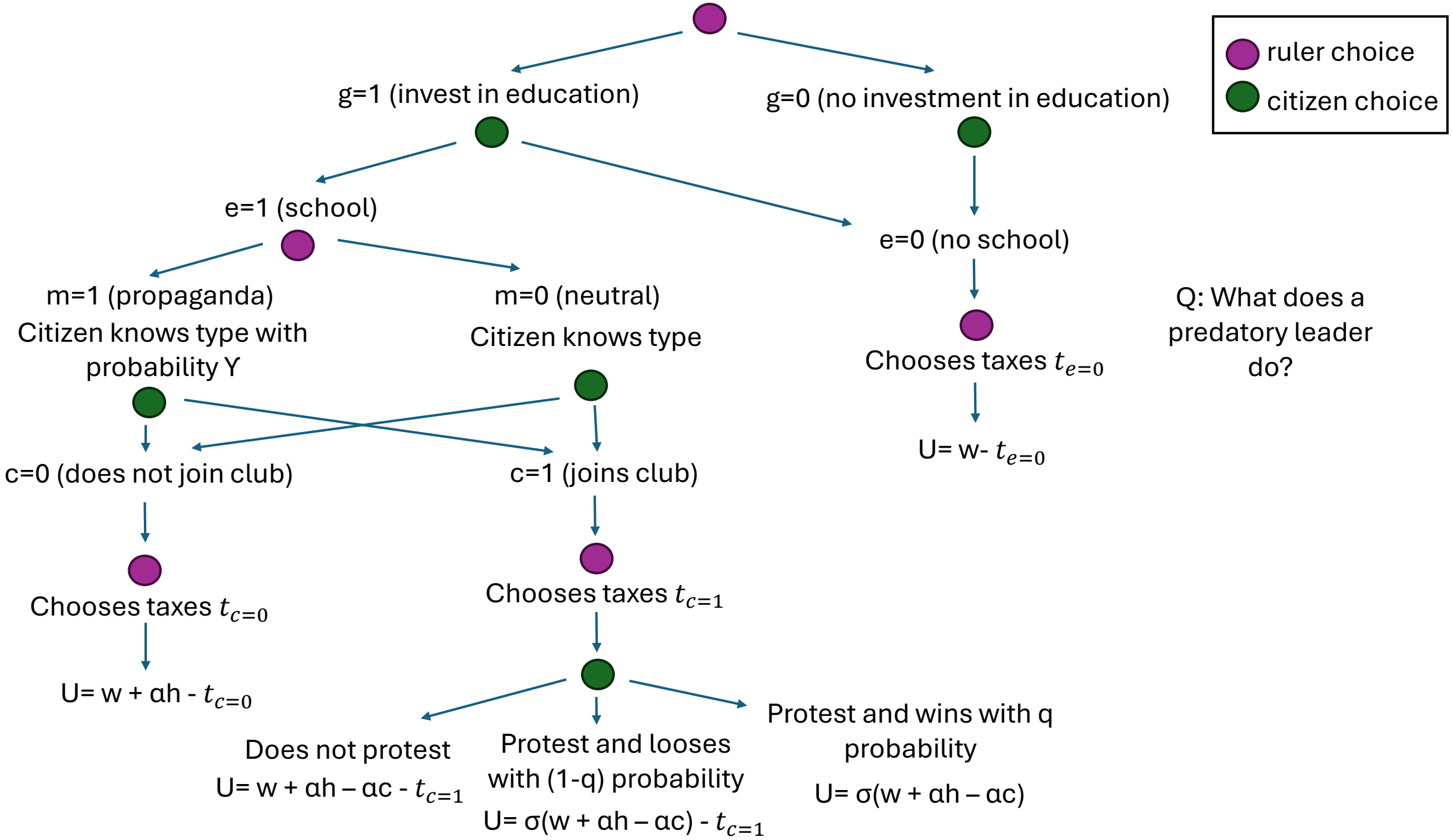


1	Does not invest in education, charge t=g
18	Invests in education, implements the neutral curriculum, and charges t=g
2	Invests in education, implements the neutral curriculum, charges t=w




ruler choice
citizen choice
g=1 (invest in education)
g=0 (no investment in education)
e=1 (school)
e=0 (no school)
m=1 (propaganda)
Citizen knows type with probability Y
m=0 (neutral)
Citizen knows type
Chooses taxes $t_{e=0}$
U= w- $t_{e=0}$
c=0 (does not join club)
c=1 (joins club)
Chooses taxes $t_{c=0}$
U= w + αh - $t_{c=0}$
Chooses taxes $t_{c=1}$
Does not protest
U= w + αh – αc - $t_{c=1}$
Protest and looses with (1-q) probability
U= σ(w + αh – αc) - $t_{c=1}$
Protest and wins with q probability
U= σ(w + αh – αc)
Q: What does a benevolent leader do?
A: $t_{c=0}$ =g (no rents)

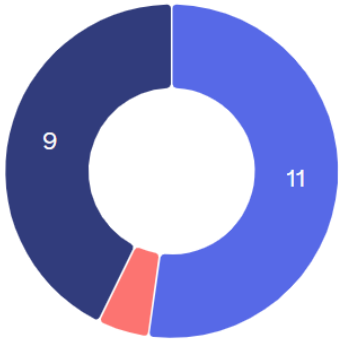

ruler choice
citizen choice
g=1 (invest in education)
g=0 (no investment in education)
e=1 (school)
e=0 (no school)
m=1 (propaganda)
Citizen knows type with probability Y
m=0 (neutral)
Citizen knows type
Chooses taxes $t_{e=0}$
U= w- $t_{e=0}$
c=0 (does not join club)
c=1 (joins club)
Chooses taxes $t_{c=0}$
U= w + αh - $t_{c=0}$
Chooses taxes $t_{c=1}$
Benevolent taxation given education provision (BE)
Does not protest
U= w + αh – αc - $t_{c=1}$
Protest and looses with (1-q) probability
U= σ(w + αh – αc) - $t_{c=1}$
Protest and wins with q probability
U= σ(w + αh – αc)
Q: What does a benevolent leader do?
A: $t_{c=0}$ =g (no rents)



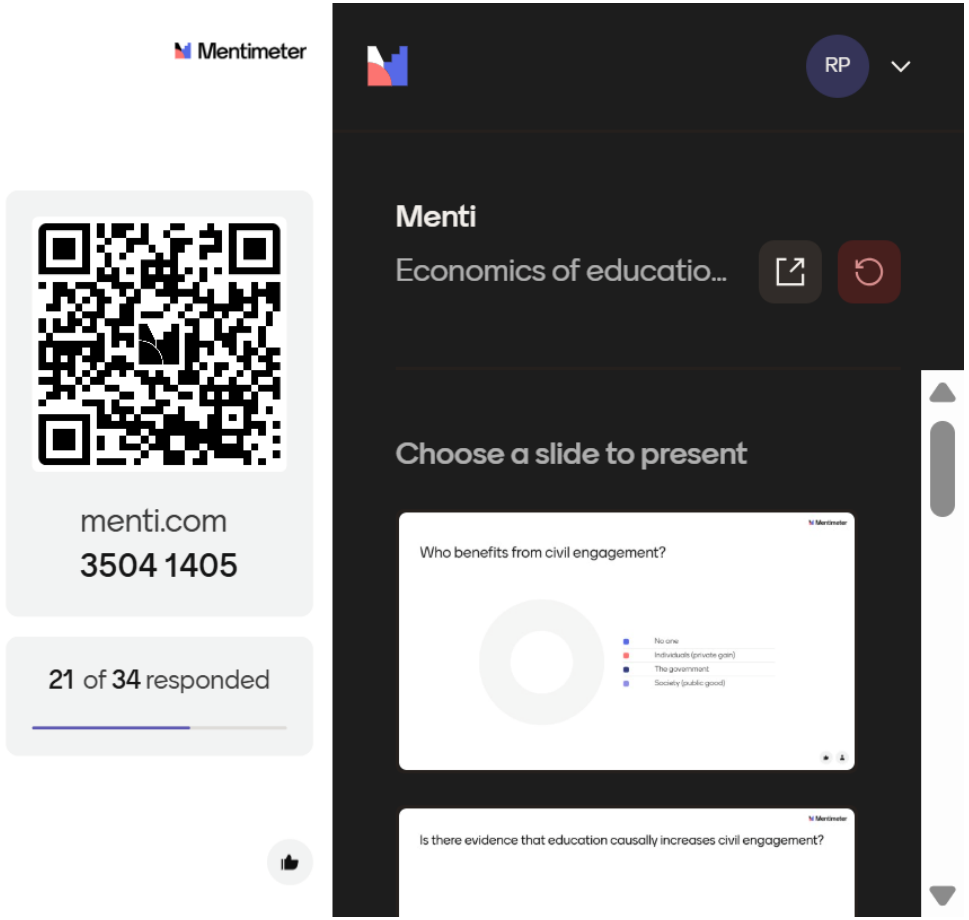
Q: What does a predatory leader do?

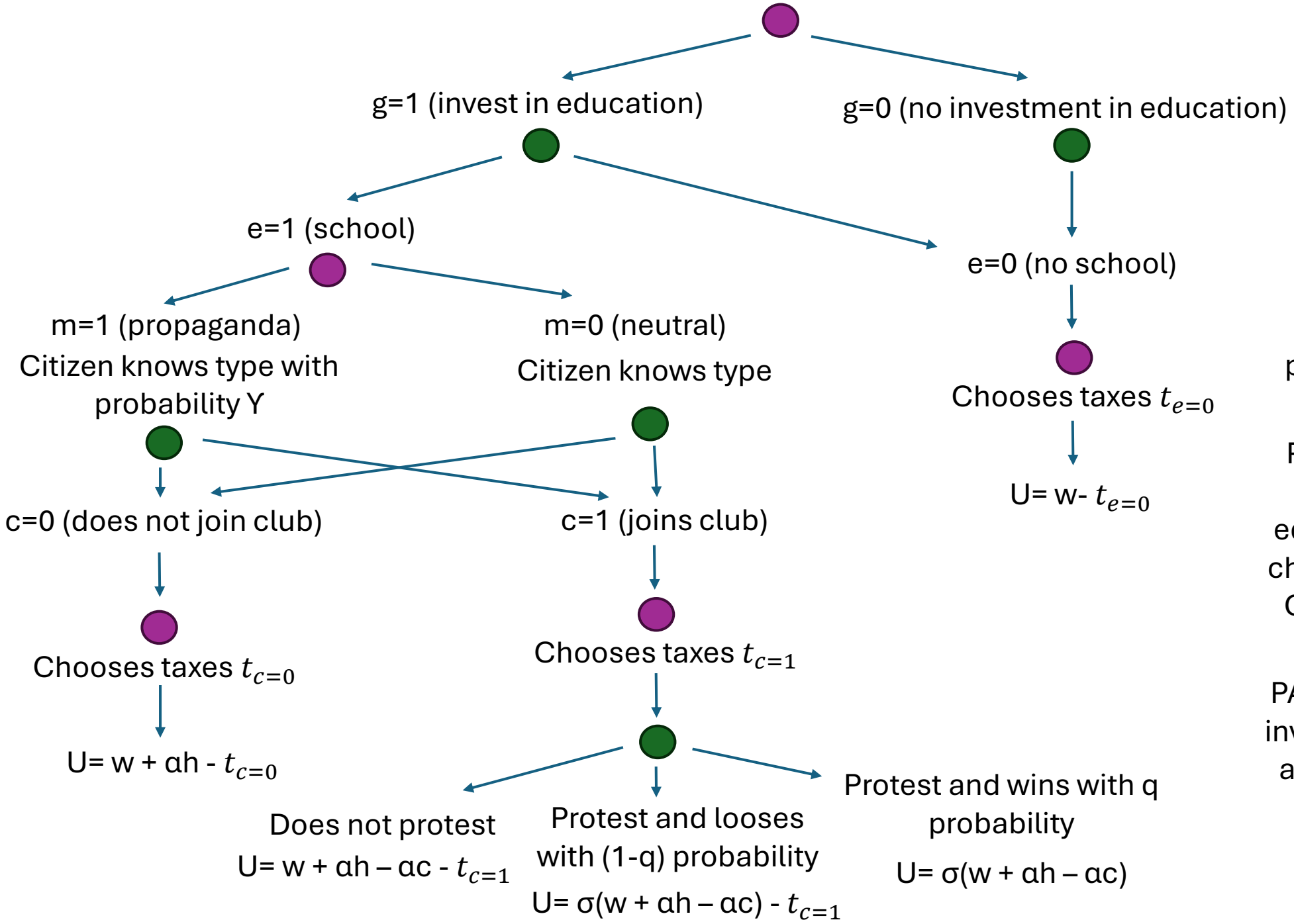
Quizz

In which node would a predatory leader want to be?



- 11 In the education, apolitical node, so he could charge t = w + αh,
- 1 In the education, political node, so he could charge t=g
- 9 In the education, political node, without protest, so he could charge t=w + αh – αc

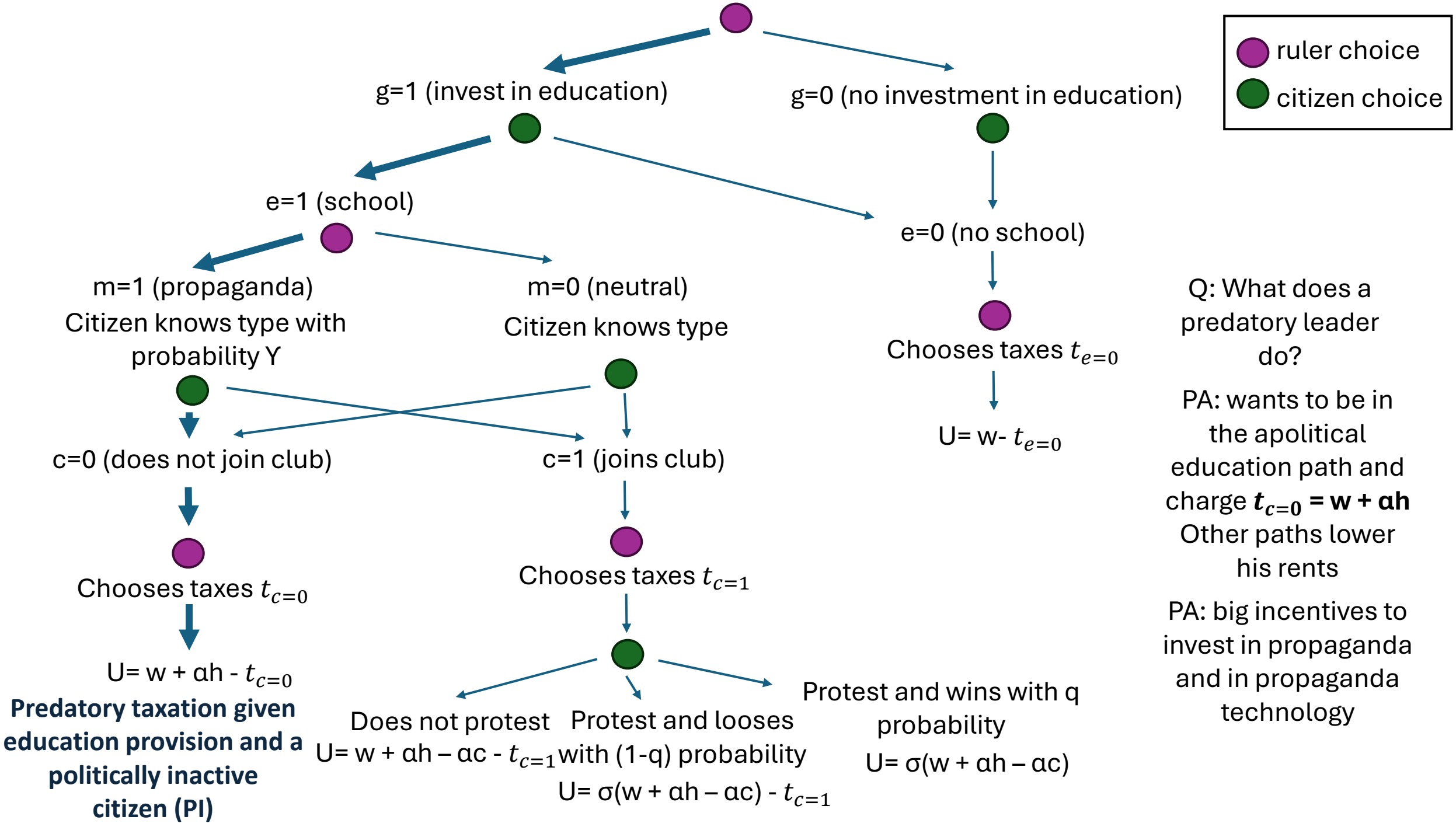




Q: What does a predatory leader do?

PA: wants to be in the apolitical education path and charge $\boldsymbol{t_{c=0}}$ **= w + αh**
Other paths lower his rents

PA: big incentives to invest in propaganda and in propaganda technology



Q: What does a predatory leader do?

PA: wants to be in the apolitical education path and charge $\boldsymbol{t_{c=0}}$ **= w + αh**
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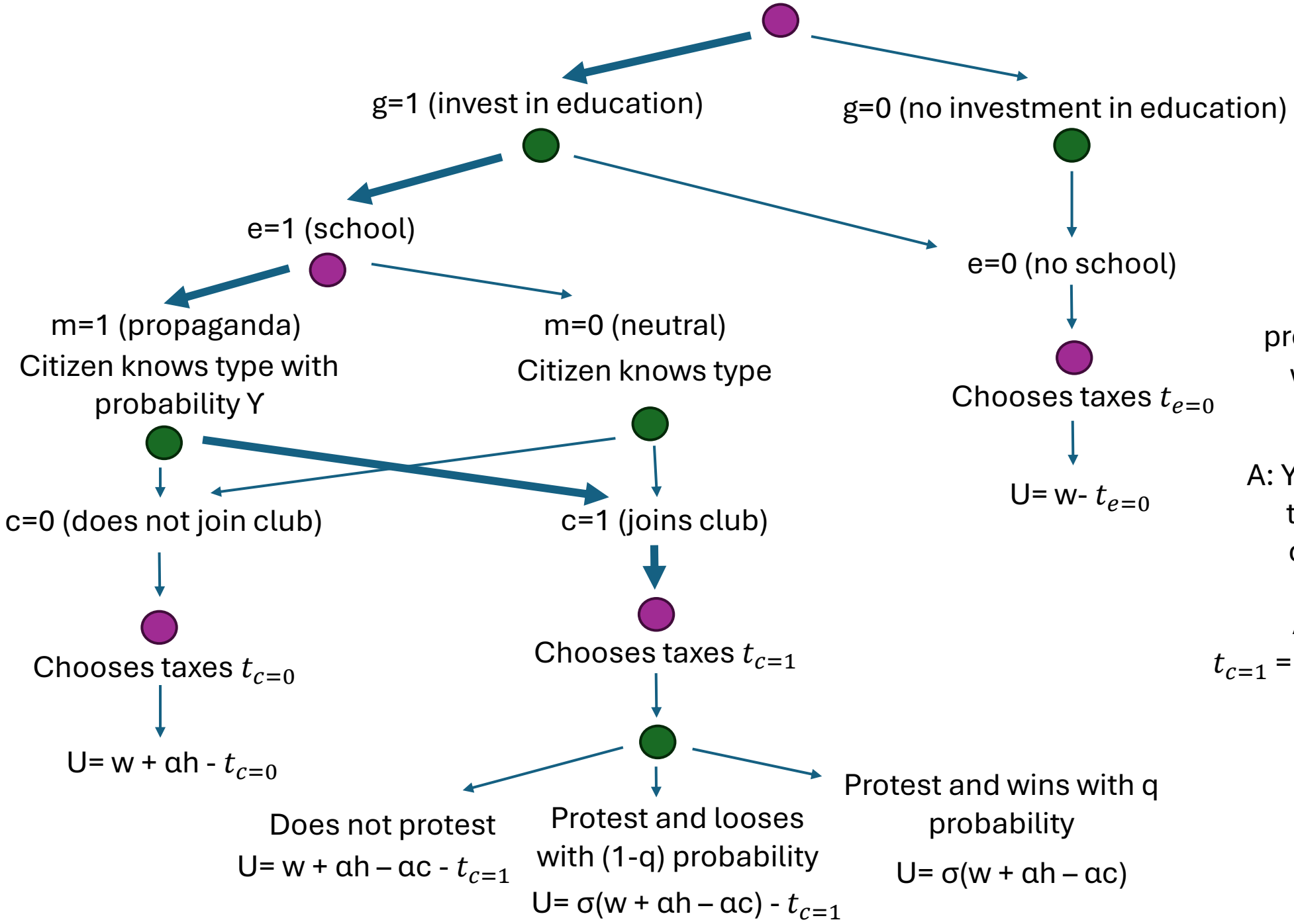


ruler choice

citizen choice

Q: What if propaganda doesn't work and citizen joins a club?

A: You maximize your tax anticipating citizens' action

A: You charge
$t_{c=1}$ = (1- qσ)(w + αh – αc)
Let's see why

proof

- $Y(e=1,c=1,h=1) = Y$ (citizens are educated and politically active)
- If protest citizen gets $q\sigma Y$
- If no protest citizen gets $Y - t$
- Indifferent between protest or not: $Y - t = q\sigma Y$, $t = (1- q\sigma)Y$

- If protest leader gets $(1-q)\sigma Y$
- To avoid protest leader gets $(1- q\sigma)Y$ (enough to make citizen indifferent)

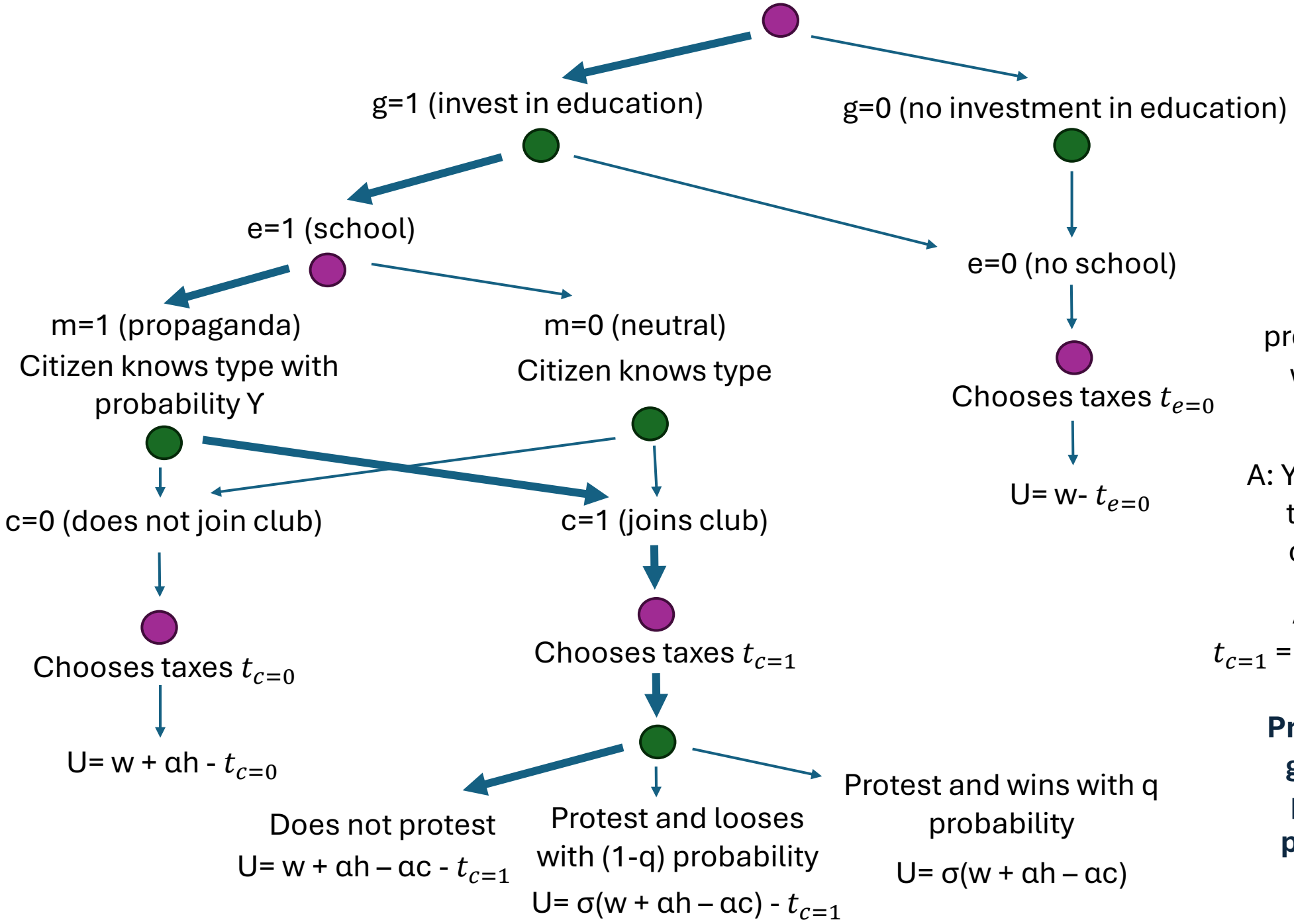
proof

- If protest leader gets $(1-q)\sigma Y$
- To avoid protest leader gets $(1- q\sigma)Y$ (enough to make citizen indifferent)

So:

$$(1- q\sigma)Y > (1-q)\sigma Y$$
$$\Leftrightarrow 1- q\sigma > (1-q)\sigma$$
$$\Leftrightarrow 1- q\sigma > \sigma - q\sigma$$
$$\Leftrightarrow 1 > \sigma$$

If there is destruction when there is a protest, then there will be a tax that will make everyone better off at that stage



Q: What if propaganda doesn't work and citizen joins a club?

A: You maximize your tax anticipating citizens' action

A: You charge
$t_{c=1}$ = (1- qσ)(w + αh – αc)

Predatory taxation given education provision and a politically active citizen (PA)

What will a predatory ruler choose?

- $t_{e=0}$ = w
- $t_{c=0}$ = w + αh
- $t_{c=1}$ = (1- qσ)(w + αh – αc)

- Leader best option is to provide education and deceive population
- Large incentives to invest in propaganda technology
- If that is not possible, then he will choose to educate if education is productive enough (αh) such that $t_{c=1} > t_{e=0}$

What will ruler choose?

- $t_{e=0}$ = w
- $t_{c=0}$ = w + αh
- $t_{c=1}$ = (1- qσ)(w + αh – αc)

- Author argues that, if propaganda doesn't work, then there is an incentive to drop it, as ruler needs to commit to lower taxations either way
- Argument: Convex effect of propaganda on beliefs?
- He makes the prediction that more productive education results in lower propaganda

Empirical analysis of the paper

- Data from the World Bank's (WB) World Development Indicators (WDI) index
- 136 countries from 1997 to 2014

Empirical analysis

- Education associates positively with propaganda proxy
- More "productive" education reduces propaganda

Table 1
Public education provision and constraints on the press and media.

Dependent variable:	Proxy for propaganda: press constraints and media censorship					
	(1a)	(1b)	(1c)	(1d)	(2)	(3)
ln*Education*	1.850**	1.850**	1.958**	1.822**	2.011**	2.517***
	(0.731)	(0.717)	(0.762)	(0.782)	(0.817)	(0.921)
ln*Edu* × ln*GDPpc*	−0.241***	−0.243***	−0.259***	−0.240**	−0.262**	−0.309***
	(0.092)	(0.091)	(0.096)	(0.098)	(0.102)	(0.116)
ln*GDPpc*	0.272	0.253	0.289	0.243	0.359	0.408
	(0.333)	(0.332)	(0.333)	(0.339)	(0.299)	(0.397)
Polity	−0.108***	−0.105***	−0.107***	−0.104***	−0.125***	−0.098***
	(0.020)	(0.020)	(0.022)	(0.023)	(0.024)	(0.026)
Crisis	–	0.176*	0.189*	0.194*	0.140	0.169
		(0.100)	(0.106)	(0.108)	(0.150)	(0.122)
ln*Military*	–	–	−0.025	0.008	0.029	0.040
			(0.102)	(0.118)	(0.096)	(0.134)
ln*GovExpend*	–	–	–	−0.045	−0.196	−0.183
				(0.215)	(0.175)	(0.237)
ln*Edu* + ln*Edu* × lnGDP_{min}	0.588**	0.578**	0.605**	0.565*	0.641**	0.872***
	(0.271)	(0.264)	(0.280)	(0.292)	(0.305)	(0.330)
ln*Edu* + ln*Edu* × ln$GDP_{10\%}$	0.363*	0.352*	0.365*	0.341	0.397*	0.586**
	(0.199)	(0.194)	(0.206)	(0.215)	(0.225)	(0.243)
ln*Edu* + ln*Edu* × ln$GDP_{90\%}$	−0.714**	−0.734**	−0.790**	−0.732**	−0.773**	−0.790*
	(0.285)	(0.287)	(0.303)	(0.304)	(0.308)	(0.374)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Balanced panel?	No	No	No	No	No	Yes
Mean imputation?	No	No	No	No	Yes	No
Observations	1452	1452	1366	1343	2134	1092
Countries	136	136	132	131	132	85
Adj. R^2	0.22	0.23	0.23	0.22	0.25	0.21

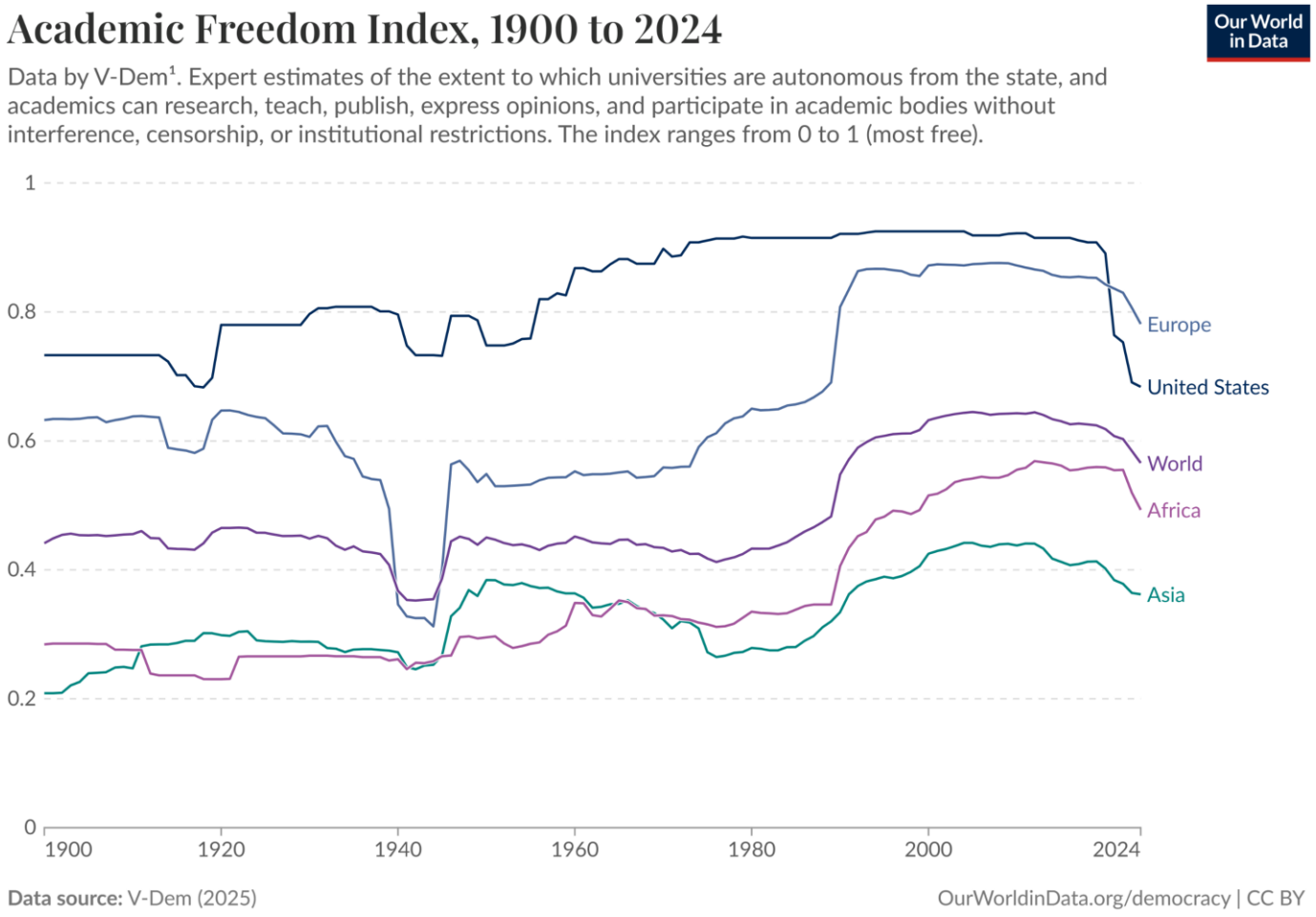
Standard errors are clustered by country, with ***, **, and * denoting significance at the 1%, 5%, and 10% levels, respectively. Column 2 applies a country mean imputation procedure to the education expenditure data and controls for a dummy, which takes a value of 1 when a value is missing. Rows labeled ln *Edu* + ln *Edu* × ln *GDP* report the combined effect (and standard errors) of ln *Education* and ln *Edu* × ln *GDPpc* at different levels (sample minimum, 10% and 90%) within the sample distribution of ln *GDPpc*.

Empirical analysis

- Model predicts a natural conversion to less propaganda
- There is a limit to how much propaganda convinces people (convex on beliefs)
- When education starts to be productive, ruler will choose to compromise taxation wise and charge the maximum possible that does not induce a protest

Reality

- Since the paper (2014) our proxy for propaganda, academic freedom as fallen pretty much everywhere in the world



1. **V-Dem** The Varieties of Democracy (V-Dem) project publishes data and research on democracy and human rights.
It relies on evaluations by around 3,500 country experts and supplementary work by its own researchers to assess political institutions and the protection of rights.
The project is managed by the V-Dem Institute, based at the University of Gothenburg in Sweden.
Learn more:
Democracy data: how do researchers measure democracy?
The 'Varieties of Democracy' data: how do researchers measure democracy?
The 'Varieties of Democracy' data: how do researchers measure human rights?

Empirical analysis

Educational attainment increases passive political participation (political club)

Table 2
Press constraints at age 18, education, and political participation, passive.

Dependent variable:	Measure of passive political participation					
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)
EducationAttainment	0.088***	0.141***	0.097**	0.183***	0.172***	0.167***
	(0.010)	(0.029)	(0.044)	(0.045)	(0.040)	(0.040)
$EA \times PressConstraints_b$	–	−0.012**	−0.008	−0.057***	−0.053***	−0.052***
		(0.005)	(0.006)	(0.017)	(0.013)	(0.012)
$EA \times PressConstraints_b^2$	–	–	–	0.005***	0.005***	0.005***
				(0.002)	(0.001)	(0.001)
$EA \times Polity_t$	–	–	0.004	0.004	0.004**	0.005**
			(0.003)	(0.003)	(0.002)	(0.002)
$PressConstraints_b$	–	0.022	0.007	0.255***	0.251***	0.258***
		(0.025)	(0.028)	(0.097)	(0.071)	(0.064)
$PressConstraints_b^2$	–	–	–	−0.027**	−0.023***	−0.024***
				(0.011)	(0.007)	(0.006)
$Polity_t$	−0.003	0.011	−0.004	−0.003	0.025	0.019
	(0.018)	(0.028)	(0.033)	(0.034)	(0.024)	(0.023)
Age_t	0.054***	0.056***	0.057***	0.055***	0.053***	0.055***
	(0.018)	(0.021)	(0.021)	(0.021)	(0.018)	(0.018)
Age_t^2	−0.001**	−0.001*	−0.001**	−0.001*	−0.001*	−0.001**
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
$IncomeLevel_t$	0.033***	0.031***	0.031***	0.032***	0.025***	0.025***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.005)	(0.006)
$Student_t$	–	–	–	–	0.158***	0.156***
					(0.030)	(0.033)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Non-students only?	Yes	Yes	Yes	Yes	No	No
Mean imputation?	No	No	No	No	No	Yes
Observations	43,714	32,432	32,432	32,432	49,552	56,129
Countries	65	65	65	65	70	70
Adj. R^2	0.13	0.13	0.13	0.13	0.11	0.11

Standard errors are clustered by country, with ***, **, and * denoting significance at the 1%, 5%, and 10% levels, respectively. Base year (b) is derived from the survey year (t), minus the respondent's age (Age_t), plus 18. All regressions control for sex, relationship status, number of children, and employment status. Columns (2c) and (2d) also apply a mean imputation procedure to educational attainment, income level, sex, relationship status, number of children, employment, and student status and control for dummies, which equal 1 when an observation is missing.

Empirical analysis

Educational attainment increases active political participation (protest)

Table 3
Press constraints at age 18, education, and political participation, active.

Dependent variable:	Measure of active political participation					
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)
EducationAttainment	0.111***	0.147***	0.115***	0.138***	0.129***	0.136***
	(0.006)	(0.016)	(0.039)	(0.043)	(0.037)	(0.033)
$EA \times PressConstraints_b$	–	−0.008***	−0.006	−0.019*	−0.019*	−0.022**
		(0.003)	(0.004)	(0.011)	(0.010)	(0.009)
$EA \times PressConstraints_b^2$	–	–	–	0.001	0.002*	0.002**
				(0.001)	(0.001)	(0.001)
$EA \times Polity_t$	–	–	0.003	0.003	0.003	0.002
			(0.003)	(0.003)	(0.002)	(0.002)
$PressConstraints_b$	–	0.033**	0.023	0.058	0.070	0.076
		(0.015)	(0.019)	(0.052)	(0.048)	(0.046)
$PressConstraints_b^2$	–	–	–	−0.004	−0.005	−0.005
				(0.005)	(0.005)	(0.004)
$Polity_t$	0.026	0.036	0.025	0.026	0.040**	0.043**
	(0.020)	(0.027)	(0.027)	(0.027)	(0.020)	(0.018)
Age_t	0.018	0.008	0.009	0.008	0.006	0.014
	(0.012)	(0.016)	(0.016)	(0.016)	(0.015)	(0.014)
Age_t^2	−0.0001	0.0001	0.0001	0.0001	0.0001	−0.00004
	(0.0002)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
$IncomeLevel_t$	0.002	0.002	0.002	0.002	0.003	0.004
	(0.006)	(0.007)	(0.007)	(0.007)	(0.005)	(0.005)
$Student_t$	–	–	–	–	0.193***	0.208***
					(0.034)	(0.040)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Non-students only?	Yes	Yes	Yes	Yes	No	No
Mean imputation?	No	No	No	No	No	Yes
Observations	41,221	30,546	30,546	30,546	46,492	52,399
Countries	65	65	65	65	70	70
Adj. R^2	0.23	0.23	0.23	0.23	0.22	0.22

Standard errors are clustered by country, with ***, **, and * denoting significance at the 1%, 5%, and 10% levels, respectively. Base year (b) is derived from the survey year (t), minus the respondent's age (Age_t), plus 18. All regressions control for sex, relationship status, number of children, and employment status. Columns (2c) and (2d) also apply a mean imputation procedure to educational attainment, income level, sex, relationship status, number of children, employment, and student status and control for dummies, which equal 1 when an observation is missing.

Take away from the model

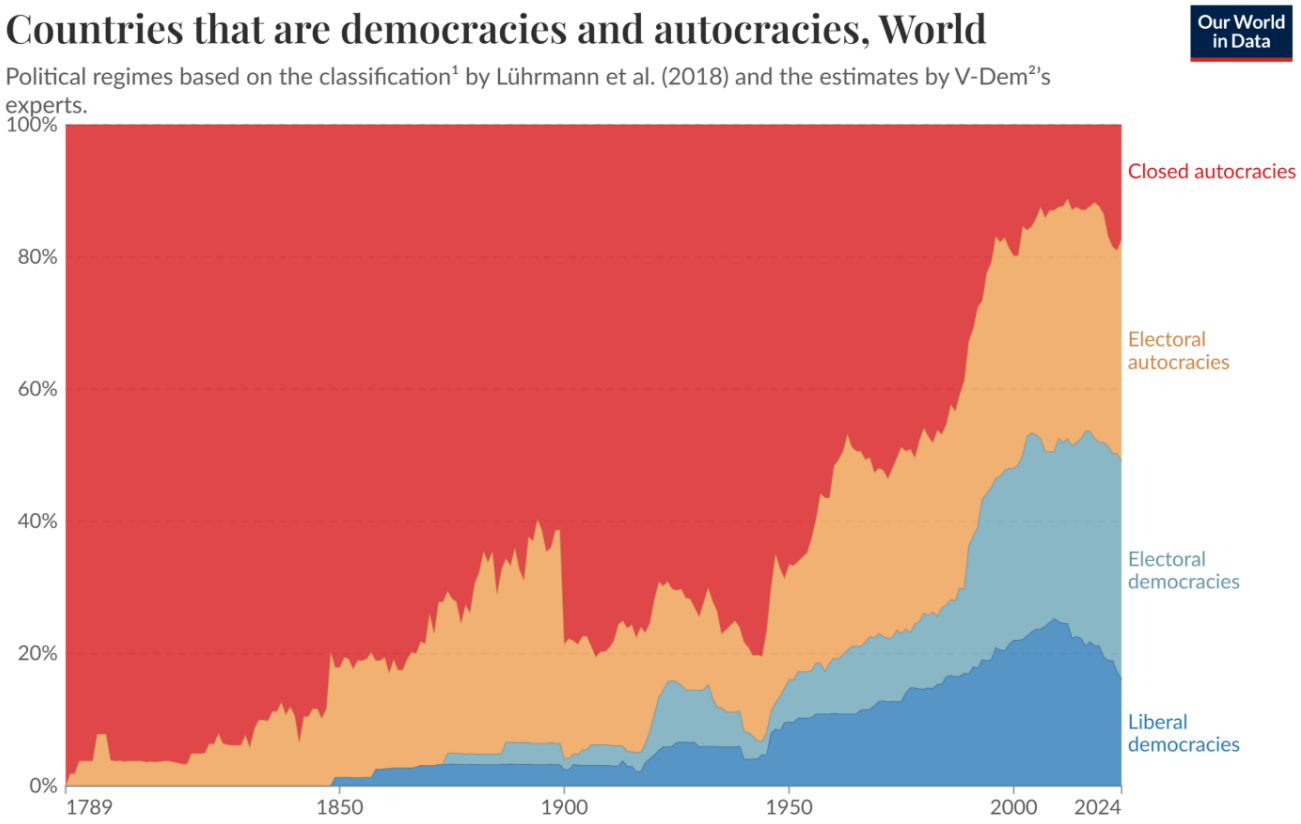
- The models builds on two empirical facts to explain behaviour of predatory/autocratic rulers:
- Education increases economic growth (more rent potential)
- Education increases political scrutiny
- A benevolent ruler collects taxes to provide education to increase economic growth
- A predatory ruler is also interested in providing education to increase economic growth and increase rents as much as possible
- In this case, an educated population can protest against the ruler

Take away from the model

- The model accurately predicts a positive association between education and passive and active political participation
- However, it also predicts a natural convergence to a non-propaganda state once education becomes sufficiently productive
- This doesn’t seem to fit latest trends

Empirical analysis

- What do you think explains latest trend?
- Propaganda technology has potentially improved?



1. Regimes of the World's regime classification: Closed autocracy: citizens do not have the right to choose either the chief executive of the government or the legislature through multi-party elections. Electoral autocracy: citizens have the right to choose the chief executive and the legislature through multi-party elections; but they lack some freedoms, such as the freedoms of association or expression that make the elections meaningful, free, and fair. Electoral democracy: citizens have the right to choose the chief executive and the legislature in meaningful, free and fair, and multi-party elections. Liberal democracy: electoral democracy and citizens enjoy individual and minority rights, are equal before the law, and the actions of the executive are constrained by the legislative and the courts. Read more in our article: The 'Regimes of the World' data: how do researchers measure democracy?

2. V-Dem: The Varieties of Democracy (V-Dem) project publishes data and research on democracy and human rights. It relies on evaluations by around 3,500 country experts and supplementary work by its own researchers to assess political institutions and the protection of rights. The project is managed by the V-Dem Institute, based at the University of Gothenburg in Sweden. Learn more: Democracy data: how do researchers measure democracy? The 'Varieties of Democracy' data: how do researchers measure democracy? The 'Varieties of Democracy' data: how do researchers measure human rights?

Take aways from the model

- However, model is useful to think about incentives and trade-offs when it comes to investing and providing education
- A predatory ruler has an incentive to invest in education if his/her propaganda technology is good (1)
- If not, it has an incentive to settle for less taxes as citizens are politically active (2)
- Some leaders might avoid education all together as they don’t want to risk losing grip over their population (3)

Take aways from the model

In general governments that:

- Disinvest in education
- Facilitate propaganda

Are likely non benevolent

The way government treats education supply and curriculum should say plenty about their character, as a good education system will increase political scrutiny (only benevolent leaders are comfortable with this)

Open questions

- What changed that can explain the trend break?
- Missing some psychology insights on predatory ruler's personalities? (maybe not rational, not willing to compromise for higher payoffs, sadism might be at play, desire for control?)
- Dynamics might change equilibrium (repeated observations over time)